

File-by-File Code Reference

Purpose and major symbols for every first-party source file.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Complete first-party file inventory

File	Purpose	Major symbols
<code>.env.example</code>	<pre># App # Generate with: python -c "import secrets; print(secrets.token_urlsafe(48))" SECRET_KEY=replace- with-a-unique-random-secret ALGORITHM=HS256 ACCESS_TOKEN_EXPIRE_MINUTES=600 ENVIRONMENT=development API_DOCS_ENABLED=true METRICS_ENABLED=true CORS_ALLOWED_ORIGINS=http://localhost:5173,http:// localhost:3000 TRUSTED_HOSTS=localhost,127.0.0.1,testserver # MongoDB MONGO_ROOT_USERNAME=replace-with- local-mongo-user MONGO_ROOT_PASSWORD=replace- with-local-mongo-password MONGO_URI=mongodb:// replace-with-local-mongo-user:replace-with-local-mongo- password@mongo:27017/?authSource=admin MONGO_DB=eurskem_ai # Weaviate WEAVIATE_HOST=weaviate WEAVIATE_PORT=8080 WEAVIATE_GRPC_PORT=50051 WEAVIATE_API_KEY_USER=eurskem-app WEAVIATE_API_KEY=replace-with-local-weaviate-api-key # MinIO MINIO_ENDPOINT=minio:9000 MINIO_ACCESS_KEY=replace-with-local-minio-user MINIO_SECRET_KEY=replace-with-local-minio-password MINIO_BUCKET=eurskem-ai-docs # Redis REDIS_PASSWORD=replace-with-local-redis-password REDIS_URL=redis://:replace-with-local-redis- password@redis:6379/0 # Grafana (local monitoring profile) GRAFANA_ADMIN_PASSWORD=replace-with- local-grafana-password # Runtime readiness.</pre>	—
<code>.github/workflows/ci.yml</code>	<pre>name: CI on: push: branches: ["main"] pull_request: permissions: contents: read concurrency: group: ci-\$ {{ github.ref }} cancel-in-progress: true jobs: backend: runs-on: ubuntu-latest timeout-minutes: 30 env: ENVIRONMENT: test SECRET_KEY: ci-only-secret-key- never-use-in-production DEV_BYPASS_ENABLED: "true" MONGO_ROOT_USERNAME: eurskem-ci MONGO_ROOT_PASSWORD: ci-mongo-only-not-a- production-secret MONGO_URI: mongodb://eurskem- ci:ci-mongo-only-not-a-production- secret@localhost:27017/?authSource=admin MONGO_DB: eurskem_ai WEAVIATE_HOST: localhost WEAVIATE_PORT: "8080" WEAVIATE_GRPC_PORT: "50051" WEAVIATE_API_KEY_USER: eurskem-ci WEAVIATE_API_KEY: ci-weaviate-only-not-a-production- secret MINIO_ENDPOINT: localhost:9000 MINIO_ACCESS_KEY: eurskem-ci MINIO_SECRET_KEY: ci-minio-only-not-a-production-secret MINIO_BUCKET: eurskem-ai-docs REDIS_PASSWORD: ci-redis-only-not- a-production-secret REDIS_URL: redis://:ci-redis-only- not-a-production-secret@localhost:6379/0 GRAFANA_ADMIN_PASSWORD: ci-grafana-only-not-a- production-secret PAPER_SEARCH_MCP_ENABLED: "false" READINESS_REQUIRED_SERVICES:</pre>	—

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	mongo,weaviate,minio,redis,checkpointner,mcp:eurskem steps: - uses: actions/ checkout@9c091bb21b7c1c1d1991bb908d89e4e9dddfe 3e0 # v7.0.0 with: persist-credentials: false - name: Install uv uses: astral-sh/setup-uv@v5 with: enable-cache: true - name: Set up Python run: uv python install 3.12 - name: Start required services run: docker compose up -d mongo weaviate minio redis - name: Wait for service health run: for attempt in {1..60}; do unhealthy="\$(docker compose ps --format json jq -rs ' [.[] if type == "array" then .[] else .	
<code>.github/workflows/deploy-ionos.yml</code>	name: Deploy IONOS Production on: workflow_run: workflows: ["CI"] types: [completed] permissions: contents: read concurrency: group: ionos-production cancel-in-progress: false jobs: deploy: if: >- github.event.workflow_run.conclusion == 'success' && github.event.workflow_run.event == 'push' && githu	—
<code>.github/workflows/live-llm-tests.yml</code>	name: Live LLM Tests on: workflow_dispatch: inputs: provider: description: Provider test to run (uses paid API calls) required: true default: openai type: choice options: - openai - anthropic - both permissions: contents: read concurrency: group: live-llm-tests cancel-in-progress: false jobs: openai	—
Dockerfile	FROM alpine:3.22 AS scientific-skills ARG SCIENTIFIC_SKILLS_REF=v2.59.0 RUN apk add --no- cache ca-certificates git \ && git clone \ --branch "\$ {SCIENTIFIC_SKILLS_REF}" \ --depth 1 \ https:// github.com/K-Dense-AI/scientific-agent-skills.git \ / scientific-agent-skills \ && rm -rf /scientific-agent-ski	—
README.md	<h1 align="center">Eurskem AI</h1> <p align="center"> An agentic workflow platform for high-stakes work. Typed workflows compiled to LangGraph, a zero-token preflight, a strict evidence lifecycle, and five canonical surfaces over one Workflow model.	—
<code>app/__init__.py</code>	Python module.	—
<code>app/api/__init__.py</code>	Python module.	—
<code>app/api/audit.py</code>	Audit module.	get_session_audit
<code>app/api/auth.py</code>	Auth module.	Token, Identity, _set_auth_cookie, login, me, logout
<code>app/api/builder.py</code>	Builder-support API — everything the visual authoring experience needs.	_services, _friendly_validation_message, operators, ContractRequest, ContractField, ContractNode, output_contract, SchemaPreviewRequest, schema_preview, NodeTestRequest, node_test, SimulateRequest
<code>app/api/candidates.py</code>	Candidates viewer API: list a run's acquired evidence PDFs with open links, plus the raw discovery/Deep Research/prior-project/official-database candidates (name + URL) recorded for the run, since ScholarlyCandidateDiscoveryAgent, BoundedDeepResearchAgent, PriorProjectRetrieverAgent,	_acquired_document_from_run, download_acquired_research_d ocument, _discovery_candidate_url, _discovered_candidates_from_r un,

File	Purpose	Major symbols
	and StructuredDatasetRetrieverAgent all produce either the same CandidateSource shape or a reconstructible exact-query URL regardless of source type.	_structured_dataset_query_url, _structured_dataset_candidates_from_run, _internal_evidence_from_run, VerifyClaimRequest, list_candidates, verify_claim_endpoint
app/api/chat_conversations.py	Durable transcript APIs for Business Chat.	_scope, _store, ResolveConversationBody, MessageBody, MessageBody.content_is_bounded, ReplaceMessageBody, resolve_conversation, get_conversation, append_message, replace_message
app/api/chat_workflows.py	Authenticated APIs for owner-scoped private Business Chat workflows.	build_deep_research_chat_workflow, _scope, _store, _slug, _private_yaml, _validate, _summary, _repair_workspace_existing_adapter, ImportChatWorkflowRequest, CopyChatWorkflowRequest, GenerateChatWorkflowRequest, _save
app/api/chat_workspace.py	Workflow-neutral planning APIs for the Chat workspace.	_managed_adapter_key, PlanWorkspaceRequest, WorkflowLabelInput, WorkflowLabelsRequest, GeneratedWorkflowLabel, GeneratedWorkflowLabels, _humanize_workflow_name, _weak_workflow_label, _workflow_label_fingerprint, _clean_workflow_label, list_workspace_experiences, chat_workflow_labels
app/api/cost.py	Cost module.	run_cost, session_cost
app/api/cost_admin.py	Account-wide cost management — the standalone admin page, distinct from app/api/cost.py's per-run/per-session views.	_db, _cost_ledger_entries, PricingOverrideRequest, list_pricing, set_pricing_override, clear_pricing_override, InfraAllocationRequest, _effective_per_call_cost, list_infra_allocations, set_infra_allocation, _allocated_infra_cost, _cache_savings_usd
app/api/email_oauth.py	Email OAuth connect flow — Outlook (Microsoft Graph) and Gmail.	ensure_indexes, _services, _require_db, _require_email_service, _connection_id, _html_result, connect, oauth_callback, AllowSendUpdate, update_connection, disconnect
app/api/entity_registry.py		

File	Purpose	Major symbols
	Manual entity-registry CRUD — Phase 1 confidential entity protection.	_scope, _service, RegisterEntityRequest, DeleteEntityRequest, list_entity_types, list_entities, register_entity, delete_entity
app/api/eval.py	Eval Lab endpoints.	RunEvalRequest, WorkflowCompareRequest, ScoreOutputRequest, _load_reference, score_output, list_golden_set, run, history, list_workflow_golden_set, workflow_compare
app/api/health.py	Liveness and dependency-readiness endpoints.	_call_probe, _probe, _mongo_probe, _weaviate_probe, _minio_probe, _redis_probe, _checkpointer_probe, probe_services, _health_payload, health, ready
app/api/inspect.py	Read-only corpus & retrieval inspection endpoints.	ChunkView, SourceGroup, ChunksResponse, RetrieveRequest, RetrievedView, RetrieveResponse, _get_weaviate, _native_collection, list_chunks, _run_retrieval, retrieve_preview
app/api/integration_oauth.py	Integration OAuth connect flow — Google Drive and OneDrive.	ensure_indexes, _services, _require_db, _require_integration_service, _connection_id, _html_result, connect, oauth_callback, list_connections, disconnect, browse_files, browse_path
app/api/knowledge.py	Knowledge Studio collection, profile, index, document and ingestion APIs.	_scope, _service, _repository, CollectionCreate, create_collection, list_collections, get_collection, ProfileCreate, create_profile, list_profiles, get_profile, create_default_profiles
app/api/llm_providers.py	Safe model-catalog and local-provider status endpoints.	_configured, list_models, list_openrouter_models, probe_model
app/api/node_types_chat.py	Ask AI about node types — explains what a node type does, when to use it, and its advantages, grounded in the live node registry so it can never describe a node type that doesn't exist and always reflects whatever node types are currently registered (no hardcoded description file to go stale).	_schema_field_names, _build_node_type_catalog, _manifest_entry, ChatMessage, AskContext, AskAboutNodeTypesRequest, _compact_catalog, ask_about_node_types, DraftPromptRequest, draft_prompt, DraftInstructionsFieldSpec, DraftInstructionsRequest
app/api/prompt_templates.py	Prompt template library APIs for Business Chat.	_scope, _store, TemplateBody, FavoriteBody,

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		list_prompt_templates, create_prompt_template, update_prompt_template, duplicate_prompt_template, favorite_prompt_template, delete_prompt_template
app/api/proposals.py	EU Proposal Evidence and Reasoning System API.	_scope, _services, _store, _graph_from_checkpoint, GraphRequest, coverage, review_run, SourceVersionRequest, register_source_version, list_source_versions, VerifyClaimsRequest, verify_claims
app/api/rag_agents.py	Saved RAG Agent management and direct test query API.	_scope, RAGAgentCreate, create_agent, list_agents, get_agent, RAGQueryRequest, query_agent
app/api/research.py	Research-feature discovery endpoints.	list_scientific_skills
app/api/retrieval.py	Retrieval Playground search, comparison and trace APIs.	presets, _scope, _service, _llm, RetrievalSearchRequest, RetrievalSearchRequest.to_query, _safe_query, search, RetrievalCompareRequest, compare, list_traces, get_trace
app/api/run_chat.py	Ask AI about a run — a grounded Q&A chat scoped to one run's own data (status, inputs, per-node outputs/errors).	starter_questions, _summarize_value, _node_configs_from_workflow_yaml, _node_type_catalog, _build_run_context, ChatMessage, AskAboutRunRequest, _scope, get_run_chat, ask_about_run
app/api/runs.py	Workflow run history, inspection, pause/resume, retry, and deletion APIs.	_scope, my_runs, my_run_detail, pending_gate, _resolve_pending_gate, _hitl_config_fallback, _hitl_display_name, RetryRunRequest, retry_failed_run, pause_run, resume_paused_run, restart_run
app/api/workflow_files.py	Authenticated multipart upload/download API for workflow input files.	_scope, capabilities, upload_workflow_input_files, ExtractWorkflowFileRequest, extract_workflow_input_file, download_workflow_input_file
app/api/workflow_generation.py	Auto-create a workflow from a natural-language prompt, tested before it's handed back.	_field_type_label, _node_type_catalog, _real_usage_snippet, _real_usage_examples, _reference_workflow_examples, _strip_code_fence, _synthesize_sample_inputs, GenerationAttempt, GenerationResult,

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		run_generation_pipeline, GenerateWorkflowRequest, build_llm_yaml_generator
app/api/workflows.py	Workflows module.	_reserve_run_id, _scope, list_workflow_root, list_node_types, allowed_models, RunRequest, ValidateWorkflowRequest, _preflight_http_detail, validate_workflow, AutofixWorkflowRequest, AutofixWorkflowResponse, AutofixNodeRequest
app/config.py	Config module.	Settings, Settings.workflow_file_max_bytes, Settings.workflow_file_max_total_bytes, Settings.max_request_body_bytes, Settings.validate_production_security, Settings.moonshot_api_key, Settings.kimi_api_base_url, Settings.required_readiness_services, Settings.resolved_paper_search_mcp_path, Settings.resolved_scientific_skills_path, Settings.allowed_scientific_skills, Settings.allowed_cors_origins
app/db/__init__.py	Python module.	—
app/db/migrations.py	Versioned, idempotent MongoDB data migrations.	MigrationError, Migration, _set_missing, _backfill_run_documents_v1, _backfill_knowledge_resources_v2, _remove_business_and_pipeline_persistence_v3, _acquire_lock, _renew_lock, run_migrations
app/db/mongo.py	Async MongoDB client and manifest collection.	MongoClient, MongoClient.ensure_client, MongoClient.manifests, MongoClient.close, MongoClient.scorecards, MongoClient.collections, MongoClient.ensure_indexes, MongoClient.upsert_manifest, MongoClient.get_manifest, MongoClient.list_manifests, MongoClient.mark_status, MongoClient.delete_manifest
app/evaluation/__init__.py	Evaluation package: LLM-as-a-Judge harness.	—
app/evaluation/golden_set.py	Golden set loading.	GoldenExample, load_golden_set

File	Purpose	Major symbols
<code>app/evaluation/judge.py</code>	LLM-as-a-Judge.	<code>_JudgeVerdict</code> , <code>LLMJudge</code> , <code>LLMJudge._score_one</code> , <code>LLMJudge.score_one</code> , <code>LLMJudge.score_all</code>
<code>app/evaluation/models.py</code>	Evaluation data models.	<code>_now_iso</code> , <code>CriterionScore</code> , <code>ExampleResult</code> , <code>ExampleResult.mean</code> , <code>Scorecard</code> , <code>Scorecard.per_criterion_mean</code>
<code>app/evaluation/runner.py</code>	Evaluation runner.	<code>_eval_one</code> , <code>run_eval</code>
<code>app/evaluation/workflow_golden.py</code>	Golden-case evaluation for a real end-to-end workflow (not a raw generation).	<code>WorkflowGoldenCase</code> , <code>load_workflow_golden_set</code> , <code>FieldCheck</code> , <code>WorkflowCaseResult</code> , <code>ModelComparisonResult</code> , <code>ModelOverrideGateway</code> , <code>ModelOverrideGateway.with_context</code> , <code>ModelOverrideGateway.complete</code> , <code>ModelOverrideGateway.complete_structured</code> , <code>ModelOverrideGateway.chat_with_tools</code> , <code>_actual_value</code> , <code>run_workflow_golden_case</code>
<code>app/evidence/__init__.py</code>	Proposal-grade evidence gathering and citation verification.	—
<code>app/evidence/claim_verification.py</code>	Best-effort secondary verification for evidence content with no URL.	<code>ClaimVerificationResult</code> , <code>_format_results</code> , <code>verify_claim</code>
<code>app/evidence/database_models.py</code>	Typed provenance contracts for structured and internal proposal evidence.	<code>StructuredDatasetQuery</code> , <code>StructuredDatasetQuery._safe_identifier</code> , <code>StructuredDatasetQuery._safe_period</code> , <code>StructuredDatasetQuery._safe_filters</code> , <code>CountReconciliation</code> , <code>StructuredDataEvidenceRecord</code> , <code>DatasetRetrievalAudit</code> , <code>InternalEvidenceRecord</code> , <code>InternalEvidenceReviewItem</code>
<code>app/evidence/identifiers.py</code>	Canonical scholarly identifier extraction and work identity.	<code>ResolvedWork</code> , <code>ResolvedWork.as_dict</code> , <code>ResolvedWork.has_strong_identifier</code> , <code>ResolvedWork.is_peer_reviewed</code> , <code>normalise_doi_value</code> , <code>_arxiv_from</code> , <code>extract_identifiers</code> , <code>normalise_title</code> , <code>work_identity</code> , <code>classify_authority</code>
<code>app/evidence/models.py</code>	Typed contracts for the EU proposal evidence factory.	<code>EvidencePolicy</code> , <code>SearchAuditRecord</code> , <code>CandidateSource</code> , <code>FullTextDocument</code> , <code>RetrievedPassage</code> , <code>ClaimEvidenceLink</code> , <code>CitationRegistryEntry</code> , <code>VerifiedClaim</code> ,

File	Purpose	Major symbols
		RejectedCandidate, EvidenceGap, EvidenceQAReport, ProposalEvidencePackage
<code>app/evidence/retrieval.py</code>	Deterministic helpers for discovery, full-text handling, and passages.	utc_now, stable_id, normalise_doi, parse_authors, parse_year, first_value, parse_mcp_payload, papers_from_payload, source_errors_from_payload, candidate_from_paper, _normalise_title_for_identity, candidate_work_identity
<code>app/ingestion/__init__.py</code>	Python module.	—
<code>app/ingestion/chunker.py</code>	Structure-aware recursive chunker.	Chunk, Chunk.embedding_content, ChunkConfig, _get_tokenizer, _split_sections, _split_paragraphs, _split_sentences, _hard_split, _pack_pieces, _split_oversized, _apply_overlap, chunk_text
<code>app/ingestion/cli.py</code>	CLI entry point for ingestion.	_parse_metadata, _main, main
<code>app/ingestion/collections.py</code>	Per-collection configuration: the controlled vocabulary that makes the platform use-case-agnostic.	CollectionConfig, CollectionConfig.validate_doc_t ype, CollectionConfig.validate_doc_t ypes, CollectionRegistry, CollectionRegistry.get, CollectionRegistry.upsert, CollectionRegistry.list_all
<code>app/ingestion/contracts.py</code>	Protocols and the stage registry for the ingestion pipeline.	DocumentParser, DocumentParser.parse, ChunkingStrategy, ChunkingStrategy.chunk, ChunkEnricher, ChunkEnricher.enrich, EmbeddingProvider, EmbeddingProvider.embed, SearchIndex, SearchIndex.index, UnknownStrategyError, StageRegistry
<code>app/ingestion/coordinator.py</code>	Starts background ingestion work and recovers unfinished jobs.	IngestionCoordinator, IngestionCoordinator.submit, IngestionCoordinator.recover, IngestionCoordinator._materiali ze_sources, IngestionCoordinator._run_safel y
<code>app/ingestion/embedder.py</code>	Embed chunks into vectors using OpenAI.	EmbedderConfig, Embedder, Embedder.embed, Embedder._embed_batch_with _retry, get_embedder
<code>app/ingestion/embedding_catalog.py</code>	Embedding models offered when building an Index.	EmbeddingModelChoice, embedding_model_catalog,

File	Purpose	Major symbols
		_looks_multilingual, select_embedding_model
app/ingestion/extractor.py	PDF, Office, plain text, and Markdown extractors.	ExtractedUnit, ExtractedDocument, ExtractedDocument.full_text, Extractor, Extractor.extract, PdfExtractor, PdfExtractor.extract, XlsxExtractor, XlsxExtractor.extract, DocxExtractor, DocxExtractor.extract, PptxExtractor
app/ingestion/indexes.py	Narrow Weaviate index writer used by ingestion.	WeaviateSearchIndex, WeaviateSearchIndex._ensure_ schema, WeaviateSearchIndex.index
app/ingestion/jobs.py	Persistent, stage-visible Knowledge Studio ingestion jobs.	IngestionCancelled, IngestionJobRunner, IngestionJobRunner._embedde r_for, IngestionJobRunner._raise_if_c ancelled, IngestionJobRunner.run
app/ingestion/pipeline.py	End-to-end document ingestion pipeline.	IngestionResult, ingest_file
app/ingestion/strategies.py	Built-in parser, chunking and enrichment strategies.	StandardDocumentParser, StandardDocumentParser.pars e, LayoutAwareDocumentParser, StructureAwareDocumentParse r, StructureAwareDocumentParse r.parse, OcrFallbackDocumentParser, OcrFallbackDocumentParser.pa rse, VisionAugmentedDocumentPar ser, VisionAugmentedDocumentPar ser.parse, _chunk_config, RecursiveChunkingStrategy, RecursiveChunkingStrategy.ch unk
app/ingestion/vision.py	Vision-model description of PDF pages carrying figures, charts and tables.	PageDescription, PdfPageVisionDescriber, PdfPageVisionDescriber.provid er, PdfPageVisionDescriber.model, PdfPageVisionDescriber.availab le, PdfPageVisionDescriber.unavail able_reason, PdfPageVisionDescriber.descri be_pages, PdfPageVisionDescriber._descri be, PdfPageVisionDescriber._build_ client, _can_render,

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		_render_pages, pages_with_visual_content
app/integrations/__init__.py	Python module.	—
app/integrations/email/__init__.py	Email integration — one capability, provider adapters underneath.	load_connections, _resolve_credentials, build_email_service
app/integrations/email/base.py	Provider-neutral email contract.	EmailAddress, EmailAddress.render, EmailAttachmentRef, EmailMessage, EmailSearchCriteria, EmailDraft, EmailResult, EmailConnection, EmailAdapterError, EmailAmbiguousFailure, EmailAdapter, EmailAdapter.search
app/integrations/email/connections_store.py	Durable, non-secret record of an OAuth-established email connection.	save_connection_record, delete_connection_record, set_allow_send, load_dynamic_connections
app/integrations/email/gmail.py	Gmail adapter — Gmail API v1 over the shared httpx client.	GmailAdapter, GmailAdapter._headers, GmailAdapter._request, GmailAdapter.search, GmailAdapter.read, GmailAdapter.create_draft, GmailAdapter.send, _gmail_query, _encode, _to_message, _extract_bodies, _extract_attachments
app/integrations/email/memory.py	In-memory email adapter.	InMemoryEmailAdapter, InMemoryEmailAdapter.add, InMemoryEmailAdapter.search, InMemoryEmailAdapter.read, InMemoryEmailAdapter.create_draft, InMemoryEmailAdapter.send
app/integrations/email/msgraph.py	Microsoft Graph adapter — Outlook / Microsoft 365 mail.	MicrosoftGraphAdapter, MicrosoftGraphAdapter._headers, MicrosoftGraphAdapter._mailbox, MicrosoftGraphAdapter._request, MicrosoftGraphAdapter.search, MicrosoftGraphAdapter.read, MicrosoftGraphAdapter.create_draft, MicrosoftGraphAdapter.send, MicrosoftGraphAdapter.reply, _graph_filters, _recipient, _draft_payload
app/integrations/email/oauth.py	OAuth authorization-code flow for Microsoft Graph and Gmail.	OAuthConfigurationError, OAuthExchangeError, TokenSet, Profile, _redirect_uri, generate_pkce_pair,

File	Purpose	Major symbols
		authorize_url, exchange_code, refresh_access_token, _parse_token_response, fetch_profile
app/integrations/email/service.py	EmailService — connection resolution, idempotency, and the operation switch.	_now, EmailConnectionError, EmailOperationInFlight, EmailOperationLedger, EmailService, EmailService.connection, EmailService.add_connection, EmailService.remove_connection, EmailService._refreshed, EmailService.adapter, EmailService.describe_connections, EmailService.execute
app/integrations/email/token_vault.py	Encrypted storage for OAuth-issued email tokens.	TokenVaultUnavailableError, _load_kek, _seal, _unseal, TokenVault, TokenVault._tokens, TokenVault.store, TokenVault.load, TokenVault.forget
app/integrations/external_action.py	ExternalActionService — the one place an outbound REST/webhook call goes through.	ExternalActionError, ExternalActionService, ExternalActionService._default_url_validator, ExternalActionService.call, ExternalActionService._replay_or_refuse, _parse_body, get_external_action_service
app/integrations/files/__init__.py	Cloud file integrations — one capability, provider adapters underneath.	load_connections, _resolve_credentials, build_integration_service
app/integrations/files/base.py	Provider-neutral cloud-file contract.	CloudFileMeta, DownloadedFile, Page, IntegrationConnection, IntegrationAdapterError, IntegrationAuthError, IntegrationNotFoundError, IntegrationProvider, IntegrationProvider.list_folder, IntegrationProvider.search_files, IntegrationProvider.get_file_metadata, IntegrationProvider.download_file
app/integrations/files/connections_store.py	Durable, non-secret record of an OAuth-established Integration connection.	save_connection_record, delete_connection_record, set_needs_reauth, load_dynamic_connections
app/integrations/files/google_drive.py	Google Drive adapter — Drive API v3 over the shared httpx client.	GoogleDriveProvider, GoogleDriveProvider._headers, GoogleDriveProvider._request, GoogleDriveProvider.list_folder, GoogleDriveProvider.search_files, GoogleDriveProvider.get_file_metadata,

File	Purpose	Major symbols
		GoogleDriveProvider.download_file, _to_meta, _to_page, _parse_timestamp
app/integrations/files/oauth.py	OAuth authorization-code flow for Google Drive and OneDrive.	OAuthConfigurationError, OAuthExchangeError, TokenSet, Profile, _redirect_uri, generate_pkce_pair, authorize_url, exchange_code, refresh_access_token, _parse_token_response, fetch_profile
app/integrations/files/onedrive.py	OneDrive adapter — Microsoft Graph over the shared httpx client.	OneDriveProvider, OneDriveProvider._headers, OneDriveProvider._request, OneDriveProvider.list_folder, OneDriveProvider.search_files, OneDriveProvider.get_file_meta, OneDriveProvider.download_file, _to_meta, _to_page, _parse_timestamp
app/integrations/files/service.py	IntegrationService — connection resolution, refresh, and the operation switch.	_now, IntegrationConnectionError, _as_id_list, IntegrationResult, IntegrationService, IntegrationService.connection, IntegrationService.add_connection, IntegrationService.remove_connection, IntegrationService._refreshed, IntegrationService.provider_for, IntegrationService.describe_connections, IntegrationService.execute
app/integrations/files/token_vault.py	Encrypted storage for OAuth-issued Google Drive / OneDrive tokens.	TokenVaultUnavailableError, _load_kek, _seal, _unseal, TokenVault, TokenVault._tokens, TokenVault.store, TokenVault.load, TokenVault.forget
app/integrations/operations.py	Durable record of attempted external side effects.	OperationInFlight, AmbiguousOperationFailure, _now, ExternalOperationLedger, ExternalOperationLedger.find, ExternalOperationLedger.reserve, ExternalOperationLedger.complete, ExternalOperationLedger.mark_ambiguous, ExternalOperationLedger.release, operation_key
app/integrations/url_guard.py	URL safety guard for ExternalActionAgent outbound calls (SSRF protection).	ExternalActionUrlError, _is_forbidden_address, validate_external_action_url

File	Purpose	Major symbols
<code>app/knowledge/__init__.py</code>	Knowledge Studio control plane: stable resources, profiles and indexes.	—
<code>app/knowledge/doc_types.py</code>	The document types a Collection can declare.	<code>DocTypeChoice</code> , <code>DocTypeChoice.precision_sensitive</code> , <code>doc_type_catalog</code>
<code>app/knowledge/ids.py</code>	Opaque, sortable Knowledge Studio resource identifiers.	<code>_encode</code> , <code>prefix_for</code> , <code>new_resource_id</code> , <code>scoped_legacy_id</code> , <code>kind_of</code> , <code>is_kind</code>
<code>app/knowledge/models.py</code>	Versioned Knowledge Studio resource contracts.	<code>utcnow</code> , <code>ResourceStatus</code> , <code>ProfileType</code> , <code>IngestionJobStatus</code> , <code>ScopedResource</code> , <code>CollectionResource</code> , <code>CollectionResource.display_name</code> , <code>DocumentResource</code> , <code>SourceVersionResource</code> , <code>ParserProfileConfig</code> , <code>ChunkingProfileConfig</code> , <code>ChunkingProfileConfig.validate_token_windows</code>
<code>app/knowledge/repository.py</code>	Owner-scoped Mongo persistence for Knowledge Studio resources.	<code>ResourceNotFoundError</code> , <code>ResourceConflictError</code> , <code>_scope</code> , <code>_strip</code> , <code>KnowledgeRepository</code> , <code>KnowledgeRepository._collections</code> , <code>KnowledgeRepository._documents</code> , <code>KnowledgeRepository._source_versions</code> , <code>KnowledgeRepository._profiles</code> , <code>KnowledgeRepository._indexes</code> , , <code>KnowledgeRepository._indexed_documents</code> , <code>KnowledgeRepository._ingestion_jobs</code>
<code>app/knowledge/service.py</code>	Authorization-aware operations over Knowledge Studio resources.	<code>validate_metadata_schema</code> , <code>workspace_for_scope</code> , <code>KnowledgeService</code> , <code>KnowledgeService.create_collection</code> , <code>KnowledgeService.create_profile_version</code> , <code>KnowledgeService.validate_profile_config</code> , <code>KnowledgeService.create_index</code> , <code>KnowledgeService.activate_index</code> , <code>KnowledgeService.create_rag_agent</code> , <code>KnowledgeService.ensure_default_profiles</code>
<code>app/llm/__init__.py</code>	LLM gateway package.	—
<code>app/llm/anthropic_batch.py</code>	Anthropic Message Batches API -- standalone, opt-in batch service.	<code>_tool_name_for</code> , <code>_build_params</code> , <code>AnthropicBatchService</code> , <code>AnthropicBatchService.submit</code> ,

File	Purpose	Major symbols
		AnthropicBatchService.status, AnthropicBatchService.cancel, AnthropicBatchService.results
app/llm/anthropic_gw.py	Anthropic Claude gateway.	_supports_temperature, _cacheable_system, _mark_last_block_cacheable, _cache_usage_fields, AnthropicGateway, AnthropicGateway.probe_model_access, AnthropicGateway._create, AnthropicGateway.complete, AnthropicGateway.complete_structured, AnthropicGateway.chat_with_tools
app/llm/base.py	Abstract LLM gateway contract.	LLMResponse, ToolCall, LLMToolUseResponse, LLMGateway, LLMGateway.complete, LLMGateway.complete_structured, LLMGateway.chat_with_tools
app/llm/batch_types.py	Provider-neutral shapes for the Anthropic/OpenAI Batch API services.	BatchRequestItem, BatchResultItem, BatchStatus
app/llm/catalog.py	Approved model catalog shared by the runtime, API, and Workflow Builder.	ModelDefinition, ModelDefinition.permits_data_class, ModelDefinition.as_dict, model_definition, provider_for_model, is_local_model, local_service_name
app/llm/errors.py	Provider-neutral LLM errors used by the resilience layer.	StructuredOutputError, LLMInputLimitError, LLMProviderUnavailableError, BatchTimeoutError, LLMPolicyDeniedError
app/llm/local_openai_gw.py	OpenAI-compatible gateway for privately hosted Kimi K3 and GLM-5.	LocalModelProfile, normalize_openai_base_url, _reasoning_content, _usage, LocalOpenAICompatibleGateway, LocalOpenAICompatibleGateway._generation_options, LocalOpenAICompatibleGateway._fixed_sampling_provider, LocalOpenAICompatibleGateway._max_tokens_field, LocalOpenAICompatibleGateway.complete, LocalOpenAICompatibleGateway.complete_structured, LocalOpenAICompatibleGateway.chat_with_tools, LocalOpenAICompatibleGateway.probe_details
app/llm/model_catalog.py	Provider-neutral LLM catalog used by routing, validation, cost, and UI.	ModelProfile, estimate_model_cost

File	Purpose	Major symbols
<code>app/llm/model_router.py</code>	Deterministic, zero-token model selection for automatic workflow routing.	ModelRoutingError, ModelSelection, ModelSelection.to_event, ModelRouter, ModelRouter.select, ModelRouter.describe_manual, infer_task_kind, infer_complexity, _request_text, _accuracy_priority, _hosted_bonus, _target_tier
<code>app/llm/openai_batch.py</code>	OpenAI Batch API -- standalone, opt-in batch service.	_build_body, OpenAIBatchService, OpenAIBatchService.submit, OpenAIBatchService.status, OpenAIBatchService.cancel, OpenAIBatchService.results, _parse_jsonl, _result_from_output_row
<code>app/llm/openai_gw.py</code>	OpenAI implementation of LLMGateway.	StructuredResult, _supports_custom_temperature, _completion_tokens_for, _cached_tokens_from_usage, _prompt_cache_key, _prompt_cache_retention_kwargs, _system_content, _system_messages, _chat_tool_reasoning_effort, OpenAIGateway, OpenAIGateway.probe_model_access, OpenAIGateway.complete
<code>app/llm/openai_registry.py</code>	Authoritative OpenAI model registry for every platform capability.	OpenAIModelDefinition, OpenAIModelDefinition.as_dict, models_for_kind, openai_model, require_openai_model_kind
<code>app/llm/openrouter_catalog.py</code>	Live OpenRouter model catalog, sourced directly from OpenRouter's own `GET /v1/models`.	OpenRouterModelInfo, OpenRouterModelInfo.as_dict, is_openrouter_model_id, _price_per_million, _to_model_info, _fetch_raw_catalog, OpenRouterCatalogCache, OpenRouterCatalogCache._is_stale, OpenRouterCatalogCache.get_models, OpenRouterCatalogCache.search, get_default_cache
<code>app/llm/openrouter_gw.py</code>	OpenRouter implementation of LLMGateway — direct calls to OpenRouter's real API.	StructuredResult, strip_openrouter_prefix, unfence_json, _parse_structured, _usage_fields, OpenRouterGateway, OpenRouterGateway.probe_model_access, OpenRouterGateway._post, OpenRouterGateway.complete,

File	Purpose	Major symbols
		OpenRouterGateway.complete_structured, OpenRouterGateway.chat_with_tools, _safe_json_loads
app/llm/openrouter_ranking.py	Auto-router config for OpenRouter — see docs/architecture/LLM_MODEL_ROUTING.md.	_cost_tier_for_priority, build_auto_router_plugin, is_openrouter_router_model
app/llm/registry.py	Resolve a model name to the right gateway instance.	_promote_tier_peers, RetryPolicy, ModelAccessResult, _status_code, _anthropic_error_type, _is_retryable_error, _provider_error_code, _is_model_unavailable_error, _retry_after_seconds, resolve_model, get_gateway, _gateway_class_for
app/llm/semantic_cache.py	Tenant-scoped Redis semantic cache for deterministic text completions.	CacheLookup, SemanticLLMCache, SemanticLLMCache.probe, SemanticLLMCache.get, SemanticLLMCache.put, SemanticLLMCache._prune, SemanticLLMCache._scope, SemanticLLMCache._entry_key, SemanticLLMCache._index_key, _cosine
app/main.py	FastAPI application entry point for Eurskem AI — Agentic Workflow Platform.	lifespan
app/mcp/business_records/__init__.py	Real, live MySQL-backed business-records MCP server.	—
app/mcp/business_records/db.py	Connection helper for the business-records MySQL database.	connect, connect_readonly, execute, execute_write
app/mcp/business_records/handlers.py	Real, parameterized-SQL handlers for the business-records MCP server.	_collection, _limit, customer_search, order_search, inventory_check, product_search, query_readonly, create_case, create_opportunity, create_order, _order_view, update_order
app/mcp/business_records/schema.sql	-- Business Records MySQL schema.	—
app/mcp/business_records/seed.py	Load both fixture files into the real MySQL database.	_load, _dt, _upsert, ensure_readonly_user, apply_schema, seed_d365_finance, seed_dynamics_crm, seed_all
app/mcp/business_records/server.py	Business Records MCP server — a real, persistent MySQL-backed connector.	backend, list_tools, call_tool, _error_result, main
app/mcp/business_records/sql_guard.py	Defense-in-depth for the query_readonly MCP tool.	SQLGuardError, classify_sql_verb, run_readonly_query, _kill_query

File	Purpose	Major symbols
<code>app/mcp/business_records/tools.py</code>	Tool definitions for the business-records MySQL server.	<code>_string</code> , <code>_collection</code> , <code>_customer_item</code> , <code>_order_item</code> , <code>_inventory_item</code> , <code>_product_item</code> , <code>_case_record</code> , <code>_opportunity_record</code> , <code>_order_record</code>
<code>app/mcp/client.py</code>	MCP client wrapper — multi-server.	<code>_paper_search_mcp_env</code> , <code>_dynamics_env</code> , <code>_finance_scm_spec</code> , <code>_business_records_env</code> , <code>build_server_specs</code> , <code>MCPCClient</code> , <code>MCPCClient.configured_servers</code> , <code>MCPCClient.running_servers</code> , <code>MCPCClient.start</code> , <code>MCPCClient.stop</code> , <code>MCPCClient._require</code> , <code>MCPCClient.has_server</code>
<code>app/mcp/connections.py</code>	Built-in MCP connections, and how the integration service is assembled.	<code>_dynamics_tool_policies</code> , <code>dynamics_connection</code> , <code>read_only_dynamics_connection</code> , <code>finance_scm_connection</code> , <code>business_records_connection</code> , <code>build_registry</code> , <code>build_mcp_service</code>
<code>app/mcp/d365_finance/__init__.py</code>	Fixture-backed mock for the d365-finance-scm-mcp business-tool layer.	—
<code>app/mcp/d365_finance/fixtures.json</code>	<pre>{ "customers": [{ "customerid": "f1a2c3d4-0000-4000-8000-000000000001", "customer_account": "C0001", "name": "Meridian Process Systems", "data_area_id": "usmf", "sales_region": "EMEA", "key_account": true, "account_owner_name": "Elena Cross", "territory_sales_owner_name": "", "sales_engineer_name":</pre>	—
<code>app/mcp/d365_finance/handlers.py</code>	Fixture-backed handlers for the d365-finance-scm-mcp business-tool layer.	<code>_collection</code> , <code>_limit</code> , <code>_any_of_present</code> , <code>_priority_filter</code> , <code>_customer</code> , <code>find_customer</code> , <code>find_account_ownership</code> , <code>find_credit_status</code> , <code>_quote</code> , <code>find_quote</code> , <code>_order_filter</code> , <code>_sales_order</code>
<code>app/mcp/d365_finance/server.py</code>	Fixture-backed mock for the Dynamics 365 Finance & Operations MCP server.	<code>backend</code> , <code>list_tools</code> , <code>call_tool</code> , <code>_error_result</code> , <code>main</code>
<code>app/mcp/d365_finance/tools.py</code>	Atomic read tools for the fixture-backed D365 Finance & SCM connector.	<code>_string</code> , <code>_collection</code>
<code>app/mcp/dynamics/__init__.py</code>	Python module.	—
<code>app/mcp/dynamics/client.py</code>	Dataverse Web API client and its fixture-backed twin.	<code>DynamicsError</code> , <code>DynamicsError.as_payload</code> , <code>DynamicsAuthError</code> , <code>DynamicsBackend</code> , <code>DynamicsBackend.whoami</code> , <code>DynamicsBackend.query</code> , <code>DynamicsBackend.get</code> , <code>DynamicsBackend.create</code> , <code>DynamicsBackend.update</code> , <code>DynamicsBackend.close</code> ,

File	Purpose	Major symbols
		DataverseClient, DataverseClient._http
app/mcp/dynamics/fixtures.json	{ "accounts": [{ "accountid": "a1b2c3d4-0000-4000-8000-000000000001", "name": "ABC Chemicals B.V.", "accountnumber": "ACC-1042", "industrycode": "Chemicals", "address1_city": "Rotterdam", "address1_country": "Netherlands", "telephone1": "+31 10 555 0142", "websiteurl": "https:// abc-chemicals.exempl	—
app/mcp/dynamics/handlers.py	Tool implementations: business question in, typed record out.	_account, _contact, _opportunity, _order, _quote, _product, _activity, _shipment, _service_case, _collection, _limit, get_current_user
app/mcp/dynamics/odata.py	Safe construction of Dataverse Web API queries.	ODataValueError, guid, escape_literal, string_filter, contains_filter, lookup_filter, any_of, all_of, entity_path, build_query, _column, _order_by
app/mcp/dynamics/server.py	The Dynamics 365 MCP server.	build_backend, backend, _annotations, list_tools, call_tool, _error_result, main
app/mcp/dynamics/tools.py	The Dynamics 365 CRM tool vocabulary.	_string, _account_summary, _contact_summary, _opportunity_summary, _product_summary, _order_summary, _activity_summary, _owner_summary, _quote_summary, _shipment_summary, _service_case_summary, _collection
app/mcp/paper_search_server.py	Credential-aware launcher for the pinned paper-search- mcp server.	_prepare_source_checkout, inject_openalex_api_key, _normalize_openaire_legacy_js on, _JsonNullSafeOpenAireRespon se, _JsonNullSafeOpenAireRespon se.json, patch_openaire_query_param, _JsonDefaultsDataKey, _JsonDefaultsDataKey.json, _wait_for_semantic_scholar_rat e_limit, patch_semantic_scholar_empty _results, main
app/mcp/policy.py	What may be called, by whom, and whether a person has to see it first.	MCPPolicyError, MCPApprovalRequired, _words, classify_by_name, classify_by_annotations, classify_tool, is_write, PolicyDecision, PolicyDecision.raise_if_denied, evaluate

File	Purpose	Major symbols
<code>app/mcp/registry.py</code>	MCP servers as first-class connections.	MCPToolPolicy, MCPServerConnection, MCPServerConnection.id_is_safe, MCPServerConnection.stdio_needs_a_command, MCPServerConnection.secrets_are_references, MCPServerConnection.label, MCPServerConnection.resolve_environment, MCPServerConnection.describe, MCPServerConnection.permits_tool, MCPServerConnection.policy_for, MCPServerRegistry, MCPServerRegistry.ids
<code>app/mcp/results.py</code>	Turn an MCP tool result into a typed workflow output.	MCPToolResult, _depth, _as_object, parse_json_text, normalise_result, _collect_text, _trim, field_paths_from_schema, _json_type
<code>app/mcp/server.py</code>	MCP server: exposes hybrid retrieval as protocol-level tools.	_services, list_tools, call_tool, main
<code>app/mcp/service.py</code>	The MCP integration service: discovery, policy, execution, audit.	MCPToolError, MCPToolError.as_payload, ToolDescriptor, MCPIntegrationService, MCPIntegrationService.describe_servers, MCPIntegrationService.discover_tools, MCPIntegrationService._describe_tool, MCPIntegrationService.find_tool, MCPIntegrationService.health_check, MCPIntegrationService.call, MCPIntegrationService._require_connection, MCPIntegrationService._replay_or_refuse
<code>app/nodes/__init__.py</code>	Built-in workflow-node discovery.	discover_nodes, node_discovery_errors
<code>app/nodes/_stubs.py</code>	Stub nodes used only to test the runtime in isolation from LLMs/RAG.	_Empty, _LiteralConfig, _LiteralOutput, LiteralNode, LiteralNode.run, _EchoConfig, _EchoOutput, EchoNode, EchoNode.run
<code>app/nodes/about_synthesis.py</code>	Fills in `about` metadata for node types that don't declare their own.	_repo_root, _edge_targets, _adjacency, _schema_fields_summary, _important_config_fields, synthesize_about

File	Purpose	Major symbols
<code>app/nodes/ai_task.py</code>	AITaskAgent — one configurable AI capability.	LanguagePolicy, LanguagePolicy.directive, AITaskExample, AITaskConfig, AITaskConfig.structured_tasks_ declare_a_schema, AITaskInput, AITaskOutput, _confidence_fields, effective_fields, AITaskAgent, AITaskAgent.required_services, AITaskAgent.preflight_output_fi elds
<code>app/nodes/approval.py</code>	Shared "did a human already approve this?" check.	human_approved
<code>app/nodes/base.py</code>	Base contract for every node type in the platform.	NodeType, NodeType.definition, NodeType.run, NodeType.required_services, NodeType.preflight_output_field s, NodeType.preflight_static_outp ut_values
<code>app/nodes/ bounded_deep_research_agent.py</code>	Run several skill-guided bounded research jobs within hard budgets.	BoundedDeepResearchInput, BoundedDeepResearchConfig, BoundedDeepResearchConfig. _coerce_briefs, BoundedDeepResearchOutput, BoundedDeepResearchAgent, BoundedDeepResearchAgent.r equired_services, BoundedDeepResearchAgent.r un, BoundedDeepResearchAgent._ record_costs, _candidate_records, _search_audit
<code>app/nodes/call_coverage.py</code>	Workflow node that computes and writes the deterministic call matrix.	CallCoverageInput, CallCoverageConfig, CallCoverageMatrixAgent, CallCoverageMatrixAgent.run
<code>app/nodes/categories.py</code>	Category taxonomy for the node palette (Builder) and Cockpit's node inspector.	category_for, icon_for, family_for, execution_kind_for
<code>app/nodes/ citation_registry_builder.py</code>	CitationRegistryBuilder: deterministic reshaping of acquired full-text documents into a renderer-shaped, numbered citation registry.	CitationRegistryBuilderInput, CitationRegistryBuilderConfig, CitationRegistryBuilderConfig._ coerce_documents, CitationEntry, CitationRegistryBuilderOutput, CitationRegistryBuilder, CitationRegistryBuilder.run
<code>app/nodes/ claim_evidence_verifier.py</code>	Verify linked proposal claims against exact, versioned source passages.	ClaimEvidenceVerifierInput, ClaimEvidenceVerifierConfig, ClaimEvidenceVerifierOutput, ClaimEvidenceVerifier, ClaimEvidenceVerifier.required_ services, ClaimEvidenceVerifier._source_t ext, ClaimEvidenceVerifier.run
<code>app/nodes/concept_alternatives.py</code>		

File	Purpose	Major symbols
	Workflow node for three grounded Horizon concept alternatives.	ConceptAlternativesInput, ConceptAlternativesConfig, ConceptAlternativesOutput, ConceptAlternativesAgent, ConceptAlternativesAgent.required_services, ConceptAlternativesAgent.run
<code>app/nodes/concept_freeze.py</code>	Deterministic freeze step for Human Gate 3 ("select and freeze the concept").	ConceptFreezeInput, ConceptFreezeConfig, ConceptFreezeConfig._coerce_alternatives, ConceptFreezeOutput, ConceptFreezeAgent, ConceptFreezeAgent.run
<code>app/nodes/consistency_checker.py</code>	ConsistencyChecker — a DETERMINISTIC node (no LLM) that gates the render.	ConsistencyCheckerOutput, ConsistencyCheckerInput, ConsistencyCheckerConfig, _check, ConsistencyChecker, ConsistencyChecker.run
<code>app/nodes/contracts.py</code>	Versioned, capability-based contracts shared by every workflow node type.	DataType, NodeConstraint, NodeDefinition, NodeDefinition.validate_contract, CompatibilityIssue, CompatibilityResult, data_types_from_schema, _collect_json_types, check_compatibility
<code>app/nodes/data_transform.py</code>	DataTransformAgent — deterministic data shaping.	TransformOperation, TransformOperation.operation_has_its_inputs, DataTransformConfig, DataTransformConfig.targets_are_unique, DataTransformInput, DataTransformOutput, DataTransformAgent, DataTransformAgent.required_services, DataTransformAgent.preflight_output_fields, DataTransformAgent.preflight_static_output_values, DataTransformAgent.run, _is_blank
<code>app/nodes/decision.py</code>	DecisionAgent — deterministic business logic, no model call.	DecisionConfig, DecisionConfig.output_field_names, DecisionInput, DecisionOutput, DecisionAgent, DecisionAgent.required_services, DecisionAgent.preflight_output_fields, DecisionAgent.preflight_static_output_values, DecisionAgent.run
<code>app/nodes/docx_renderer.py</code>	<code>app/nodes/docx_renderer.py</code> DOCX tool node — pre-baked variant: - DOCXProposalRenderer: render sections to a styled .docx The .docx twin of PDFProposalRenderer.	DOCXRenderInput, DOCXRenderConfig, DOCXRenderOutput, DOCXProposalRenderer,

File	Purpose	Major symbols
		DOCXProposalRenderer.require d_services, DOCXProposalRenderer.run, DOCXProposalRenderer._rende r_docx_bytes, DOCXProposalRenderer._add_t itle_block, DOCXProposalRenderer._style_ heading, DOCXProposalRenderer._rende r_markdown, DOCXProposalRenderer._rende r_table, DOCXProposalRenderer._add_i nline
app/nodes/dynamic_figure_agent.py	DynamicFigureAgent: generate an image for every inline image marker and embed it as a data URI the DOCX renderer can actually render.	DynamicFigureAgentInput, DynamicFigureAgentConfig, GeneratedFigure, DynamicFigureAgentOutput, DynamicFigureAgent, DynamicFigureAgent.run, _sub_captioned, _escape
app/nodes/email_integration.py	EmailAgent — one email capability with an operation selector.	EmailRecipient, EmailNodeConfig, EmailNodeConfig.operation_ha s_what_it_needs, EmailNodeInput, EmailMessageSummary, EmailNodeOutput, EmailAgent, EmailAgent.required_services, EmailAgent.preflight_output_fiel ds, EmailAgent.run, _criteria, _draft
app/nodes/end.py	EndAgent: the canonical workflow exit point.	EndOutputField, EndConfig, EndInput, EndOutput, _humanize, EndAgent, EndAgent.required_services, EndAgent.preflight_output_field s, EndAgent.run
app/nodes/evidence_agent.py	Scholarly candidate discovery for proposal claims.	ClaimSearchPlan, _sanitize_query, ScholarlyCandidateDiscoveryIn put, ScholarlyCandidateDiscoveryC onfig, ScholarlyCandidateDiscoveryO utput, _ScholarlyCandidateDiscovery, _ScholarlyCandidateDiscovery.r equired_services, _ScholarlyCandidateDiscovery.r un, _ScholarlyCandidateDiscovery. _query_planning_guidance, _ScholarlyCandidateDiscovery. _search_plan, ScholarlyCandidateDiscoveryA gent
app/nodes/excel_tool.py		ExcelInput, ExcelConfig, ExcelOutput,

File	Purpose	Major symbols
	Excel tool node — pre-baked variant: extract tables from .xlsx in MinIO.	ExcelTableExtractor, ExcelTableExtractor.required_services, ExcelTableExtractor.run
app/nodes/external_action.py	ExternalActionAgent — call a REST API or a webhook, kept apart from MCP.	ExternalActionConfig, ExternalActionInput, ExternalActionOutput, ExternalActionAgent, ExternalActionAgent.required_services, ExternalActionAgent.run, ExternalActionAgent._failure
app/nodes/figure_embedder.py	Splice generated figures into compiled proposal text before rendering.	FigureSpec, FigureEmbedderInput, FigureEmbedderConfig, FigureEmbedderOutput, _marker_pattern, FigureEmbedder, FigureEmbedder.run
app/nodes/graph_normalizer.py	GraphNormalizer — turns source text into VALIDATED typed ProposalGraph objects.	GraphNormalizerOutput, GraphNormalizerInput, GraphNormalizerConfig, GraphExtraction, GraphNormalizer, GraphNormalizer.required_services, GraphNormalizer.run, GraphNormalizer._parse_json
app/nodes/horizon_docx_renderer.py	Horizon proposal HTML/Markdown to editable DOCX workflow node.	HorizonDOCXProposalRenderInput, HorizonDOCXProposalRenderConfig, HorizonDOCXProposalRenderOutput, _safe_component, HorizonDOCXProposalRender, HorizonDOCXProposalRender.required_services, HorizonDOCXProposalRender.run
app/nodes/horizon_evaluation.py	Workflow node for independent two-provider Horizon evaluation.	HorizonEvaluationInput, HorizonEvaluationConfig, HorizonEvaluationAgent, HorizonEvaluationAgent.required_services, HorizonEvaluationAgent.run
app/nodes/html_proposal_renderer.py	Horizon proposal HTML/Markdown to PDF workflow node.	HorizonHTMLProposalRenderInput, HorizonHTMLProposalRenderConfig, HorizonHTMLProposalRenderOutput, _safe_component, HorizonHTMLProposalRender, HorizonHTMLProposalRender.required_services, HorizonHTMLProposalRender.run
app/nodes/human_in_loop.py	HumanInLoopAgent: pause and wait for user decision.	ReviewPanel, HITLConfig, HITLInput, HITLReviewContent,

File	Purpose	Major symbols
		HITLOutput, ReviewPanelValue, HITLInterruptPayload, HumanInLoopAgent, HumanInLoopAgent.required_services, HumanInLoopAgent.run, _resolve_path, _review_panels
app/nodes/integration.py	IntegrationAgent — one cloud-storage capability with a provider selector.	IntegrationNodeConfig, IntegrationNodeConfig.operation_has_what_it_needs, IntegrationNodeInput, CloudFileMetaOut, IntegrationNodeOutput, IntegrationAgent, IntegrationAgent.required_services, IntegrationAgent.preflight_output_fields, IntegrationAgent.run, _meta_out, _to_workflow_file_ref, _error_output
app/nodes/internal_project_evidence_retriever.py	Retrieve exact, approval-gated facts from internal project sources.	_InternalFactDraft, _InternalExtraction, InternalProjectEvidenceRetrieverInput, InternalProjectEvidenceRetrieverConfig, InternalProjectEvidenceRetrieverOutput, InternalProjectEvidenceRetrieverAgent, InternalProjectEvidenceRetrieverAgent.required_services, InternalProjectEvidenceRetrieverAgent.run, _research_questions, _normalise_registry, _source_blocks, _source_class
app/nodes/kimi_vision_agent.py	Understand one uploaded image with Kimi K3 vision.	KimiVisionInput, KimiVisionConfig, KimiVisionOutput, KimiVisionAgent, KimiVisionAgent.required_services, KimiVisionAgent.run
app/nodes/knowledge_retrieval.py	One generic retrieval-only workflow capability.	KnowledgeRetrievalError, KnowledgeRetrievalConfig, KnowledgeRetrievalInput, KnowledgeRetrievalOutput, KnowledgeRetrieval, KnowledgeRetrieval.required_services, KnowledgeRetrieval.run, KnowledgeRetrieval._validate_resources, KnowledgeRetrieval._retrieve_with_policy
app/nodes/mcp_agent.py	MCPAgent: an LLM-driven agent loop calling MCP tools.	MCPAgentConfig, MCPAgentInput, ToolCallRecord, MCPAgentOutput, MCPAgent,

File	Purpose	Major symbols
		MCPAgent.required_services, MCPAgent.run
app/nodes/mcp_tool.py	MCPToolAgent — call one capability on one MCP server.	MCPToolConfig, MCPToolInput, MCPToolOutput, MCPToolAgent, MCPToolAgent.required_services, MCPToolAgent.preflight_output_fields, MCPToolAgent.run, MCPToolAgent._required_arguments, MCPToolAgent._skipped, MCPToolAgent._failure, _empty_arguments, _summarise
app/nodes/ methodology_engineering.py	Produce one structured Method Card per frozen-concept objective.	MethodologyEngineeringInput, MethodologyEngineeringConfig, _MethodCardDraft, MethodCard, MethodologyEngineeringOutput, MethodologyEngineeringAgent, MethodologyEngineeringAgent.required_services, MethodologyEngineeringAgent.run, MethodologyEngineeringAgent._draft_one_card
app/nodes/ minio_evidence_ingestion.py	MinIOEvidenceIngestion: index already-acquired full-text sources into Weaviate so the drafters can be GROUNDED in the papers, not merely cite them.	MinIOEvidenceIngestionInput, MinIOEvidenceIngestionConfig, MinIOEvidenceIngestionConfig._reject_unresolved, MinIOEvidenceIngestionOutput, MinIOEvidenceIngestion, MinIOEvidenceIngestion.run, _load_pages, _source_path, _chunk_pages, _rfc3339_now
app/nodes/ openai_image_generation_agent.py	Generate one OpenAI image and persist it in object storage.	ImageGenerationInput, ImageGenerationConfig, ImageGenerationOutput, OpenAllImageGenerationAgent, OpenAllImageGenerationAgent.required_services, OpenAllImageGenerationAgent.run
app/nodes/ paperqa_evidence_synthesizer.py	Per-document, gap-aware literature synthesis over already-acquired sources.	PaperQAEvidenceSynthesizerInput, PaperQAEvidenceSynthesizerConfig, PaperQAEvidenceSynthesizerConfig._coerce_documents, DocumentCoverage, ClaimSynthesis, PaperQAEvidenceSynthesizerOutput, PaperQAEvidenceSynthesizerAgent, PaperQAEvidenceSynthesizerAgent.required_services, PaperQAEvidenceSynthesizerA

File	Purpose	Major symbols
		gent.run, PaperQAEvidenceSynthesizerA gent._synthesize_claim
app/nodes/pdf_tool.py	PDF tool node — two pre-baked variants: - PDFTextExtractor: read .pdf, return text per page - PDFProposalRenderer: render sections to styled .pdf	PDFExtractInput, PDFExtractConfig, PDFExtractOutput, PDFTextExtractor, PDFTextExtractor.required_serv ices, PDFTextExtractor.run, PDFRenderInput, PDFRenderConfig, PDFRenderOutput, PDFProposalRenderer, PDFProposalRenderer.required _services, PDFProposalRenderer.run
app/nodes/powerpoint_tool.py	PowerPoint tool node — pre-baked variant: generate slide deck from sections.	PPTInput, PPTConfig, PPTOutput, PowerPointProposalSlides, PowerPointProposalSlides.requ ired_services, PowerPointProposalSlides.run
app/nodes/ prior_project_retriever.py	Official prior-project discovery for Horizon proposal positioning.	PriorProjectRetrieverInput, PriorProjectRetrieverConfig, PriorProjectRetrieverConfig._co erce_briefs, PriorProjectRetrieverOutput, PriorProjectRetrieverAgent, PriorProjectRetrieverAgent.requ ired_services, PriorProjectRetrieverAgent.run, _allowed_official_url
app/nodes/ proposal_evidence_factory.py	Full-text, claim-level evidence verification for EU proposals.	PairVerdict, ProposalEvidenceFactoryInput, ProposalEvidenceFactoryConfi g, ProposalEvidenceFactoryConfi g._coerce_candidates, ProposalEvidenceFactoryConfi g._coerce_documents, ProposalEvidenceFactoryConfi g._coerce_search_audit, ProposalEvidenceFactoryConfi g._coerce_rejected_candidates, ProposalEvidenceFactoryOutput, _materiality, _authority, _strength, _evidence_score
app/nodes/ proposal_submission_gate.py	Deterministic final release gate for proposal-engineering workflows.	ProposalSubmissionGateInput, ProposalSubmissionGateConfi g, ProposalSubmissionGateOutput, _messages, ProposalSubmissionGate, ProposalSubmissionGate.run
app/nodes/proposal_truth_graph.py	Create the drafting-safe, integrity-stamped proposal truth graph.	ProposalTruthGraphInput, ProposalTruthGraphConfig, ProposalTruthGraphConfig._co erce_verified_claims, ProposalTruthGraphConfig._co

File	Purpose	Major symbols
		erces_evidence_gaps, ProposalTruthGraphConfig._coerce_structured_data_records, ProposalTruthGraphConfig._coerce_internal_evidence_records, ProposalTruthGraphOutput, ProposalTruthGraphAgent, ProposalTruthGraphAgent.run, _review_queue, _string_list
app/nodes/python_snippet.py	PythonSnippetAgent — write-and-run a Python snippet as a workflow step.	PythonSnippetConfig, PythonSnippetInput, PythonSnippetOutput, PythonSnippetAgent, PythonSnippetAgent.required_services, PythonSnippetAgent.preflight_static_output_values, PythonSnippetAgent.run, PythonSnippetAgent._failure, scan_snippet_for_warnings, assigns_output
app/nodes/rag.py	RAGAgent: hybrid retrieval + grounded generation with citations.	RAGConfig, RAGConfig.reject_ambiguous_saved_agent_config, RAGInput, Citation, RAGOutput, RAGAgent, RAGAgent.required_services, RAGAgent.run
app/nodes/registry.py	Global registry of node types.	_EmptySchema, NodeRegistryError, NodeRegistry, NodeRegistry.register, NodeRegistry.get, NodeRegistry.manifest, _manifest_entry, _uses_llm, _with_model_catalog
app/nodes/research_source_acquirer.py	Acquire Deep Research citations directly, without a paper-search MCP.	ResearchSourceAcquirerInput, ResearchSourceAcquirerConfig, ResearchSourceAcquirerConfig._coerce_candidates, ResearchSourceAcquirerOutput, ResearchSourceAcquirer, ResearchSourceAcquirer.required_services, ResearchSourceAcquirer.run, _acquire_candidate, _bounded_candidates, _fetch_public_source, _require_public_url, _html_pages
app/nodes/router.py	RouterAgent: the workflow's branching primitive.	RouteRule, RouteCase, RouteCase.case_is_decidable, RouterConfig, RouterConfig.mode_has_what_it_needs, RouterConfig.multi_selection_not_supported_in_rule_mode, RouterConfig.route_names, RouterOutput, RouterInput,

File	Purpose	Major symbols
		RouterAgent, RouterAgent.required_services, RouterAgent.run
<code>app/nodes/scientific_research_planner.py</code>	Proposal-aware research planning guided by curated scientific skills.	_ResearchBriefDraft, _ResearchPlanDraft, ScientificResearchPlannerInput, ScientificResearchPlannerConfig, ScientificResearchPlannerOutput, ScientificResearchPlannerAgent, ScientificResearchPlannerAgent.required_services, ScientificResearchPlannerAgent.run, _json_text
<code>app/nodes/scientific_skill_agent.py</code>	LLM research node guided by approved K-Dense Scientific Agent Skills.	ScientificSkillConfig, ScientificSkillInput, ScientificSkillOutput, ScientificSkillAgent, ScientificSkillAgent.required_services, ScientificSkillAgent.run
<code>app/nodes/sql_query.py</code>	SQLQueryAgent — a read-only SQL lookup, for a query the classified MCP tools don't already cover.	SQLQueryConfig, SQLQueryInput, SQLQueryOutput, SQLQueryAgent, SQLQueryAgent.required_services, SQLQueryAgent.preflight_output_fields, SQLQueryAgent.run, SQLQueryAgent._failure
<code>app/nodes/start.py</code>	StartAgent: the canonical workflow entry point.	StartFileField, StartFileField.resolved_source, StartConfig, StartInput, StartOutput, StartAgent, StartAgent.required_services, StartAgent.preflight_output_fields, StartAgent.preflight_static_output_values, StartAgent.run, _as_field_specs
<code>app/nodes/structured_dataset_retriever.py</code>	Reproducible retrieval of official structured proposal evidence.	_StructuredDatasetPlan, StructuredDatasetRetrieverInput, StructuredDatasetRetrieverConfig, StructuredDatasetRetrieverConfig._coerce_queries, StructuredDatasetRetrieverOutput, StructuredDatasetRetrieverAgent, StructuredDatasetRetrieverAgent.required_services, StructuredDatasetRetrieverAgent.run, StructuredDatasetRetrieverAgent._plan_queries, _json_text,

File	Purpose	Major symbols
		<code>_deduplicate_queries</code> , <code>_decode_jsonstat</code>
<code>app/nodes/subprocess.py</code>	SubprocessAgent — call another saved workflow as a reusable business subprocess.	SubprocessConfig, SubprocessInput, SubprocessOutput, SubprocessAgent, SubprocessAgent.required_services, SubprocessAgent.run, SubprocessAgent._get_or_create_launch, SubprocessAgent._finalize, _load_child_workflow, _resolve_child_inputs, _coerce_child_input
<code>app/nodes/text_assembler.py</code>	TextAssemblerAgent: this platform's business-facing Join node.	TextAssemblerConfig, TextAssemblerInput, TextAssemblerOutput, TextAssemblerAgent, TextAssemblerAgent.run
<code>app/nodes/transform.py</code>	TransformAgent: pure LLM transform with structured output.	TransformInputField, TransformConfig, TransformConfig.has_a_prompt_source, TransformConfig.deterministic_mode_has_operations, is_new_style, _render_input_value, _is_semantically_empty, TransformInput, TransformOutput, _decode_json_container, _build_response_model, TransformAgent
<code>app/nodes/web_search_agent.py</code>	Typed workflow node for live public-web candidate discovery.	WebSearchAgentInput, WebSearchAgentConfig, WebSearchHit, WebSearchAgentOutput, WebSearchAgent, WebSearchAgent.required_services, WebSearchAgent.run
<code>app/nodes/workflow_file_loader.py</code>	Node that turns uploaded workflow-file references into usable content.	WorkflowFileLoaderInput, WorkflowFileLoaderConfig, LoadedWorkflowFile, WorkflowFileLoaderOutput, WorkflowFileLoader, WorkflowFileLoader.required_services, WorkflowFileLoader.run
<code>app/nodes/workflow_input.py</code>	WorkflowInputAgent — the entry primitive.	_friendly_number, InputFieldBinding, InputFieldBinding._default_incomplete_legacy_shape, InputFieldBinding.resolved_source, WorkflowInputConfig, WorkflowInputConfig.as_field_specs, WorkflowInputInput, WorkflowInputOutput, WorkflowInputAgent, WorkflowInputAgent.required_services,

File	Purpose	Major symbols
		WorkflowInputAgent.preflight_output_fields, WorkflowInputAgent.preflight_static_output_values
app/observability/__init__.py	Python module.	—
app/observability/cost_ledger.py	Cost ledger module.	LedgerEntry, configure_pricing_db, _refresh_pricing_overrides_if_stale, CostLedger, CostLedger.calculate, CostLedger.record, CostLedger.record_no_charge, CostLedger._group_by, CostLedger.run_summary, CostLedger.session_summary, CostLedger.query
app/observability/logging.py	Logging module.	configure_logging, get_logger
app/observability/metrics.py	Prometheus metrics for the agentic workflow platform.	track_node, record_paused, record_llm_usage
app/observability/tracing.py	Optional OpenTelemetry wiring.	configure_tracing
app/proposal_graph/__init__.py	Proposal workspace graph: domain state and workspace store.	—
app/proposal_graph/concepts.py	Generate three independently-produced concept alternatives, then score and adversarially judge each one separately from its generation.	ConceptAlternativeSet, ConceptAlternativeSet.one_of_each_posture, _ConceptDraft, _ConceptJudgment, _valid_ids, score_concept, _composite_score, _draft_one_concept, judge_concept, generate_concept_alternatives
app/proposal_graph/coverage.py	Deterministic Horizon call-coverage matrix.	CallCoverageRow, CallCoverageMatrix, _known_object_ids, build_call_coverage_matrix
app/proposal_graph/evidence_verification.py	Claim-evidence verification with an exact-passage safety check.	ClaimSupportVerdict, _normalise_space, quote_exists_in_source, verify_claim_against_text, relation_from_verdict
app/proposal_graph/graph.py	ProposalGraph — the container that lives inside WorkflowState under one key.	ProposalGraph, _merge_object, merge_graph, empty_graph
app/proposal_graph/horizon_evaluator.py	Independent two-provider Horizon Europe proposal evaluation panel.	HorizonJudgeVerdict, HorizonJudgeResult, HorizonCriterionPanel, HorizonEvaluationReport, _provider, validate_independent_models, deterministic_horizon_blockers, evaluate_horizon_proposal
app/proposal_graph/models.py	Proposal Knowledge Graph — typed objects.	Status, Authority, EvidenceStance, ConceptPosture, CallRequirement,

File	Purpose	Major symbols
		EvidenceSource, Claim, EvidenceRelation, Objective, Innovation, Result, Outcome
app/proposal_graph/state.py	Workflow-state adapter for the EU proposal use-case pack.	proposal_graph_from_state, proposal_graph_state_update
app/proposal_graph/workspace_store.py	Immutable proposal snapshots, source versions, and approval decisions.	_now, _canonical_hash, SourceVersionRecord, ProposalSnapshot, ApprovalRecord, ProposalWorkspaceStore, ProposalWorkspaceStore.ensure_indexes, ProposalWorkspaceStore.register_source_version, ProposalWorkspaceStore.list_source_versions, ProposalWorkspaceStore.source_text, ProposalWorkspaceStore.save_snapshot, ProposalWorkspaceStore.request_approval
app/rag/__init__.py	Saved RAG Agent resources and grounded generation.	—
app/rag/models.py	Saved RAG Agent query request/response contracts.	RAGCitation, RAGQueryResponse
app/rag/service.py	Resolve a saved RAG Agent, retrieve securely, then generate an answer.	_build_sources, _build_relevant_context, RAGService, RAGService._select_route, RAGService.query
app/research/__init__.py	Controlled adapters for external scientific research assets.	—
app/research/deep_research.py	Bounded Deep Research contracts and provider adapter.	_projected_call_cost, ResearchBrief, ResearchCitation, ResearchToolTrace, ResearchUsage, ResearchDossier, BoundedToolResearchService, BoundedToolResearchService.available, BoundedToolResearchService.research, BoundedToolResearchService._attribute_citations, _CitationAttribution, _CitationAttributionSet
app/research/skills.py	Allowlisted adapter for K-Dense Scientific Agent Skills.	SkillDocument, SkillDocument.metadata, SkillSelection, SkillSelection.names, SkillSelection.versions, ScientificSkillCatalog, ScientificSkillCatalog.refresh, ScientificSkillCatalog.probe, ScientificSkillCatalog.metadata, ScientificSkillCatalog.load_errors,

File	Purpose	Major symbols
		ScientificSkillCatalog.loaded_s kill_names, ScientificSkillCatalog.select
app/retrieval/__init__.py	Hybrid RAG retrieval pipeline.	—
app/retrieval/compressor.py	Stage 4 of the retrieval pipeline: contextual compression.	_compress_one, compress_chunks
app/retrieval/context.py	Parent/sentence-window/contextual expansion and final context assembly.	GraphContextProvider, GraphContextProvider.expand, expand_context, assemble_context
app/retrieval/contracts.py	Small interfaces for retrieval, fusion, reranking and context expansion.	RetrievalStrategy, FusionStrategy, Reranker, ContextExpander
app/retrieval/filters.py	Typed metadata filters and the non-removable authorization-filter clauses.	metadata_property_name, MetadataFilterValidationError, _field_type, _check_field, _check_value, validate_metadata_document, _validate_group, validate_metadata_filters, coerce_metadata_filter_group, _predicate_to_filter, _group_to_filter, build_secure_where_filter
app/retrieval/fusion.py	Reciprocal Rank Fusion for independently executed retrieval result sets.	reciprocal_rank_fusion, deduplicate
app/retrieval/hybrid_search.py	Stage 2 of the retrieval pipeline: hybrid search with metadata pre-filter.	_build_where_filter, hybrid_search
app/retrieval/models.py	Pydantic models for the retrieval pipeline.	MetadataFilterPredicate, MetadataFilterGroup, RetrievalFilters, RetrievalFilters.user_filter_dum p, RetrievalQuery, RetrievedChunk, RetrievalStage, RetrievalResult
app/retrieval/presets.py	Editable Retrieval Playground starting configurations.	—
app/retrieval/query_transform.py	Rewrite, multi-query, decomposition, HyDE and validated self-query.	_safe_complete, _split_lines, transform_query
app/retrieval/ query_understanding.py	Stage 1 of the retrieval pipeline: query understanding.	rewrite_query
app/retrieval/reranker.py	Stage 3 of the retrieval pipeline: LLM-as-judge reranking.	_RerankScore, _RerankResponse, _format_candidates, rerank
app/retrieval/retriever.py	The retrieval pipeline orchestrator.	retrieve
app/retrieval/service.py	Canonical retrieval path for workflows, RAG, Playground, MCP and Eval.	RetrievalAuthorizationError, RetrievalCompatibilityError, _safe_candidate, RetrievalService, RetrievalService.retrieve, RetrievalService._record_failed _trace,

File	Purpose	Major symbols
		RetrievalService._retrieve_impl, RetrievalService.compare
app/retrieval/strategies.py	Replaceable dense, sparse and hybrid Weaviate retrieval strategies.	_chunk_from_object, dense_search, sparse_search, native_hybrid_search, fetch_chunks_by_id
app/retrieval/weaviate_client.py	Weaviate client and schema definition.	ensure_collection_schema_on, upsert_objects_on, WeaviateClient, WeaviateClient.connect, WeaviateClient.close, WeaviateClient.ensure_schema, WeaviateClient.ensure_collection_schema, WeaviateClient.upsert_objects, WeaviateClient.reset_schema, WeaviateClient.upsert_chunks, WeaviateClient.count_chunks, WeaviateClient.add_property
app/runtime/__init__.py	Phase 4 runtime: YAML to LangGraph compiler and executor.	—
app/runtime/autofix.py	Auto-fix preflight errors.	DeterministicFixResult, _get_path, _set_path, _replace_template_reference, _replace_template_node_id, _node_by_id, _fix_template_unknown_output_field, _fix_template_unknown_structured_field, _fix_template_unknown_node, _replace_template_named_root_segment, _fix_template_unknown_named_root, _fix_template_unknown_input
app/runtime/compiler.py	Compile a WorkflowSpec into a runnable LangGraph StateGraph.	_make_runtime_fn, _wire_edges, compile_workflow
app/runtime/coordination.py	Small Redis coordination primitives shared by Uvicorn workers.	RedisLease, RedisLease.acquire, RedisLease.renew, RedisLease.release, RedisLease.keep_alive
app/runtime/domain_state.py	Namespaced state for optional use-case packs.	_merge_mapping, DomainStateRegistry, DomainStateRegistry.register, DomainStateRegistry.reducer_for, merge_domain_state
app/runtime/events.py	Bounded, session-scoped pub/sub for workflow Server-Sent Events.	RunEvent, RunEvent.terminal, RunEvent.to_json, RunEventBus, RunEventBus._key, RunEventBus.publish, RunEventBus._redis_names, RunEventBus._publish_redis, RunEventBus.subscribe, RunEventBus._subscribe_redis,

File	Purpose	Major symbols
		RunEventBus._decode_redis_event, RunEventBus._offer
app/runtime/executor.py	Compile a WorkflowSpec and run it.	_find_rejection, _project_output, run_workflow
app/runtime/field_schema.py	Visual field schema — the authoring format behind the Builder's structured output builder.	FieldSpec, FieldSpec._migrate_legacy_item_enum_values, FieldSpec.name_is_an_identifier, FieldSpec.shape_is_coherent, validate_fields, _scalar_annotation, _annotation_for, _build_model, _safe_model_name, build_response_model, json_schema, FieldPath
app/runtime/hitl.py	HITL resume path.	HITLResumeError, HITLResumeConflict, _find_rejection, _resume_output, resume_workflow, _validate_saved_decision, resume_workflow_durable
app/runtime/loader.py	Loader module.	load_workflow, load_workflow_from_string
app/runtime/logic_preflight.py	Zero-token validation of authored business logic.	_register_builders, _declared_value, _infer_kind, _json_schema_kind, _json_schema_paths, _static_mcp_tool_registries, _static_mcp_tool_definition, _static_mcp_output_schema, _static_mcp_required_arguments, _static_subprocess_child_spec, _shallow_child_node_paths, _transform_kind
app/runtime/node_events.py	Helpers for live event emission in the workflow runtime.	sanitize_preview, is_graph_interrupt, interrupt_payload
app/runtime/preflight.py	Zero-token workflow validation.	DuplicateYamlKeyError, _UniqueKeyLoader, _construct_unique_mapping, PreflightSeverity, PreflightIssue, PreflightCheck, WorkflowPreflightReport, WorkflowPreflightReport.errors, WorkflowPreflightReport.warnings, WorkflowPreflightReport.refresh, WorkflowPreflightError, _issue
app/runtime/rules.py	Deterministic business-rule evaluation.	operators_for_type, resolve_path, path_exists, _resolve, Condition, Condition.value_present_when_needed, ConditionGroup, ConditionGroup.group_is_usable, Action,

File	Purpose	Major symbols
		Action._rename_legacy_appended, Action.value_matches_operation, Rule
app/runtime/schema.py	Schema module.	WorkflowFileRef, WorkflowInputSpec, WorkflowInputSpec.validate_file_options, WorkflowInputSpec.effective_max_files, ModelRoutingPolicy, GuidedStageSpec, WorkflowExperienceSpec, NodeExperienceSpec, LibraryDurationRange, LibraryHumanReviews, LibraryEvidencePolicy, LibraryMetadataSpec
app/runtime/sleep_guard.py	macOS-only guard against idle sleep while a workflow is executing.	_enabled, acquire, release
app/runtime/snippet_client.py	App-side client for the snippet-runner sidecar (app/runtime/snippet_daemon.py).	SnippetRunnerUnavailable, SnippetRunnerClient, SnippetRunnerClient.run, SnippetRunnerClient.probe, SnippetRunnerClient._call
app/runtime/snippet_daemon.py	The snippet-runner sidecar's own entrypoint.	_socket_path, _clamped, _rlimit_preexec, _execute, _handle, main
app/runtime/snippet_protocol.py	Shared shape between the app and the snippet-runner sidecar.	read_message, SnippetRequest, SnippetResponse
app/runtime/state.py	Shared mutable state that flows through every LangGraph node.	merge_node_outputs, WorkflowState
app/runtime/templating.py	Resolve {{node_id.field.subfield}} expressions inside node configs.	_lookup, _resolve_reference, _as_text, resolve, _union_members, _accepts_none, _element_annotation, prune_absent
app/security/__init__.py	Python module.	—
app/security/audit.py	Audit log writer + reader.	_require_session, summarize_payload, write_audit_event, read_audit_events
app/security/dependencies.py	Dependencies module.	CurrentUser, _user_from_token, get_current_user, get_current_user_cookie_or_header, require_permission
app/security/entity_ner.py	Lazy-loaded spaCy NER safety net — PERSON/ORG spans not already known.	EntityMatch, _load_pipeline, extract_entities
app/security/entity_protection_errors.py	Fail-closed exception hierarchy for confidential entity protection.	EntityProtectionError, EntityVaultUnavailableError, EntityScanUnavailableError, VaultKeyMisconfiguredError,

File	Purpose	Major symbols
		UnresolvedEntityRiskError, ResponseLeakDetectedError
<code>app/security/entity_registry.py</code>	Explicit per-scope entity registry — the primary detection mechanism.	EntityRegistry, EntityRegistry.register, EntityRegistry.list_entities, EntityRegistry.delete, EntityRegistry.sync_from_partners
<code>app/security/entity_tokenizer.py</code>	Entity tokenization orchestration — the Phase 1 core guarantee.	ProcessingMode, TokenizeResult, DetokenizeResult, _Span, _overlaps, _boundary_pattern, _ambiguous_short_ner_match, EntityTokenizerService, EntityTokenizerService.registry, EntityTokenizerService.ensure_indexes, EntityTokenizerService.tokenize, EntityTokenizerService._detect_spans
<code>app/security/entity_vault.py</code>	Encrypted, reversible placeholder<->original-value vault.	_load_kek, _normalize, entity_hash, _mapping_id, _counter_id, _expiry, _seal, _unseal, EntityVault, EntityVault._mappings, EntityVault._counters, EntityVault.ensure_indexes
<code>app/security/guardrails.py</code>	Deterministic input/output guardrails for workflow execution.	GuardrailViolation, GuardrailFinding, GuardrailResult, _redact_matches, _walk, check_workflow_inputs, check_generated_output
<code>app/security/jwt_handler.py</code>	Jwt handler module.	create_access_token, decode_token
<code>app/security/llm_policy.py</code>	Data classification + OPA policy enforcement for direct provider calls.	processing_mode_to_data_class, effective_data_class, provider_and_backend_for_gateway, ProviderComplianceMetadata, _parse_csv_env, get_provider_compliance_metadata, PolicyDecision, OpaClient, OpaClient.evaluate, get_default_opa_client, enforce_policy
<code>app/security/middleware.py</code>	HTTP request identity, limits, security headers, and Redis rate limiting.	RequestContextMiddleware, RequestContextMiddleware.dispatch, RequestSizeLimitMiddleware, RequestSizeLimitMiddleware.dispatch, RedisRateLimitMiddleware, RedisRateLimitMiddleware.dispatch, _rate_limit_subject, _route_label

File	Purpose	Major symbols
<code>app/security/passwords.py</code>	Argon2 password hashing helpers for local production authentication.	<code>hash_password</code> , <code>verify_password</code>
<code>app/security/presidio_client.py</code>	Presidio Analyzer client — the PII/entity detection safety-net tier.	<code>PresidioMatch</code> , <code>_map_entity_type</code> , <code>analyze</code>
<code>app/security/rbac.py</code>	Rbac module.	<code>Role</code> , <code>has_permission</code>
<code>app/security/sso_stub.py</code>	Microsoft SSO stub.	<code>stub_login</code>
<code>app/security/users.py</code>	Mongo-backed local users used by the IONOS production deployment.	<code>ensure_user_indexes</code> , <code>validate_username</code> , <code>upsert_local_user</code> , <code>authenticate_local_user</code>
<code>app/storage/__init__.py</code>	Python module.	—
<code>app/storage/minio_client.py</code>	S3-compatible object storage client.	<code>content_hash</code> , <code>key_for_path</code> , <code>knowledge_key_for_path</code> , <code>ObjectStore</code> , <code>ObjectStore.ensure_bucket</code> , <code>ObjectStore.put_file</code> , <code>ObjectStore.put_bytes</code> , <code>ObjectStore.get_bytes</code> , <code>ObjectStore.object_exists</code> , <code>ObjectStore.list_objects</code> , <code>ObjectStore.presigned_url</code> , <code>get_object_store</code>
<code>app/tools/content_length.py</code>	Character-count helper for the proposal-size guardrails.	<code>narrative_char_count</code>
<code>app/tools/database_lookup.py</code>	Bounded clients for documented public database APIs.	<code>DatabaseResponse</code> , <code>PublicDatabaseLookupService</code> , <code>PublicDatabaseLookupService</code> . <code>close</code> , <code>PublicDatabaseLookupService</code> . <code>query_eurostat</code> , <code>PublicDatabaseLookupService</code> . <code>_client</code> , <code>PublicDatabaseLookupService</code> . <code>_rate_limit</code> , <code>_pairs_for_audit</code> , <code>get_database_lookup_service</code>
<code>app/tools/docx_proposal_rendering.py</code>	Secure Horizon proposal HTML/Markdown to editable DOCX rendering.	<code>ProposalDocxRenderResult</code> , <code>_InlineStyle</code> , <code>_TablePlacement</code> , <code>_BuildState</code> , <code>_set_run_font</code> , <code>_set_shading</code> , <code>_set_run_shading</code> , <code>_append_styled_text</code> , <code>_set_paragraph_borders</code> , <code>_set_cell_shading</code> , <code>_set_cell_margins</code> , <code>_set_cell_width</code>
<code>app/tools/excel_io.py</code>	Read tables from an Excel file.	<code>read_tables_from_xlsx</code>
<code>app/tools/image_io.py</code>	Provider-neutral image generation used by workflow nodes.	<code>GeneratedImage</code> , <code>OpenAllImageGenerationService</code> , <code>OpenAllImageGenerationService</code> . <code>available</code> , <code>OpenAllImageGenerationService</code> . <code>generate</code> , <code>OpenAllImageGenerationService</code> . <code>_generate_openrouter</code> , <code>_image_request_kwargs</code> ,

File	Purpose	Major symbols
		_model_unavailable, get_image_generation_service
app/tools/pdf_io.py	PDF read + render utilities.	extract_text_from_pdf, render_proposal_pdf
app/tools/powerpoint_io.py	Build a proposal .pptx from a dict of {section: text}.	build_proposal_pptx
app/tools/proposal_rendering.py	Secure, proposal-aware HTML/Markdown to PDF rendering.	ProposalRenderResult, _markdown, _slug, _unique_id, _sanitise_fragment, _toc, _safe_url_fetcher, render_horizon_proposal
app/tools/templates/corporate.css	@page { size: A4; margin: 2cm; @bottom-right { content: counter(page); } } body { font-family: "Liberation Sans", Helvetica, sans-serif; color: #1a2b4a; } .cover h1 { font-size: 32pt; color: #002b5c; border-bottom: 4px solid #002b5c; padding-bottom: 12pt; } .cover .client { font-size: 14pt; color: #	—
app/tools/templates/horizon_part_b.css	@page { size: A4; margin: 18mm 17mm 19mm 20mm; @top-left { content: element(page-header); font-family: Arial, Helvetica, sans-serif; font-size: 8pt; color: #4b5563; } @bottom-left { content: element(page-footer); font-family: Arial, Helvetica, sans-serif; font-size: 8pt; color: #4b5563; } @bottom-ri	—
app/tools/templates/professional.css	/* =====	—
	professional.css — EU Horizon Europe Work Programme style Modelled on the Horizon Europe Work Programme 2026-2027 (Health) document: A4, Times New Roman, EU blue, running header + "Part / Page X of Y" footer.	
app/tools/templates/warm.css	@page { size: A4; margin: 2cm; } body { font-family: "Liberation Sans", Verdana, sans-serif; color: #3a2a1a; background: #fbf6ef; } .cover h1 { font-size: 30pt; color: #b04a2f; } .cover .client { font-size: 13pt; color: #6b4a30; } .proposal-section { page-break-before: always; } .proposal-section h2	—
app/tools/vision_io.py	Kimi K3 vision service for image understanding.	VisionAnalysis, KimiVisionService, KimiVisionService.available, KimiVisionService.analyze, get_kimi_vision_service
app/tools/web_io.py	Live web-search providers with an auditable fallback policy.	WebResult, WebSearchResponse, TavilyQuotaExhausted, WebSearchService, WebSearchService.available_pro viders, WebSearchService.resolve_pro vider, WebSearchService.search, WebSearchService._select_pro vider, WebSearchService._search_tav ily, WebSearchService._search_op enai, WebSearchService._search_ki mi,

File	Purpose	Major symbols
		WebSearchService._stub_results
app/workflow/builder_store.py	Durable, file-backed state used by the Workflow Builder.	_utc_now, _digest, _atomic_write, WorkflowBuilderStore, WorkflowBuilderStore.validate_name, WorkflowBuilderStore.validate_version_id, WorkflowBuilderStore.workflow_path, WorkflowBuilderStore.draft_path, WorkflowBuilderStore.versions_dir, WorkflowBuilderStore.save_draft, WorkflowBuilderStore.read_draft, WorkflowBuilderStore.delete_draft
app/workflow/capability_selection.py	Deterministic, zero-token capability → node-type shortlisting for workflow generation (see app/api/workflow_generation.py).	_split_pascal_case, _tokens, _entry_text, select_candidate_node_types, select_generation_model
app/workflow/chat_conversation_store.py	Durable, owner-scoped Business Chat conversations and transcript messages.	ChatConversationNotFoundError, ChatConversationRecord, ChatConversationRecord.public, ChatConversationMessageRecord, ChatConversationMessageRecord.public, ensure_chat_conversation_indexes, ChatConversationStore, ChatConversationStore.get_or_create, ChatConversationStore.get, ChatConversationStore.list_messages, ChatConversationStore.append_message, ChatConversationStore.replace_message
app/workflow/chat_output_compatibility.py	Static compatibility analysis for Business Chat's visible output contract.	ChatOutputCompatibility, _terminal_ids, analyze_chat_output
app/workflow/chat_workflow_store.py	Owner-scoped persistence for private Business Chat workflows.	_default_output_compatibility, ChatWorkflowConflictError, ChatWorkflowNotFoundError, ChatWorkflowRecord, ChatWorkflowRecord.public_summary, ensure_chat_workflow_indexes, ChatWorkflowStore, ChatWorkflowStore.create, ChatWorkflowStore.list_private, ChatWorkflowStore.get, ChatWorkflowStore.get_by_slug,

File	Purpose	Major symbols
		ChatWorkflowStore.get_by_adapter_key
app/workflow/chat_workspace_planner.py	Deterministic planning for the workflow-powered Chat workspace.	WorkspaceExperience, WorkspaceExperience.public, WorkspacePlan, WorkspacePlan.public, _experience, _base, _experience_block, _answer_node, _reply, build_existing_workflow_chat_adapter, build_llm_adapter, build_scientific_skill_adapter
app/workflow/claim_verifications.py	Durable storage for claim-verification results (see app.evidence.claim_verification).	_key, ensure_indexes, get_verifications, save_verification
app/workflow/file_inputs.py	Workflow file-input contracts, validation, and object-storage helpers.	WorkflowFileInputError, scope_token, workflow_input_prefix, safe_filename, extension_for, category_for_extension, content_type_for, accepted_by_spec, store_upload, validate_workflow_inputs, validate_workflow_file_reference, extract_workflow_file_text
app/workflow/library_metadata.py	Derives Workflow Library catalog metadata and readiness from a WorkflowSpec.	_humanize, _terminal_node_types, infer_output_types, _default_human_review_count, library_summary, readiness_summary
app/workflow/orchestration.py	Shared execution bookkeeping for a single workflow run.	start_new_run_record, record_run_failure, finalize_run_result, run_and_finalize, BackgroundRunManager, BackgroundRunManager.launch, BackgroundRunManager._run_owned, BackgroundRunManager.close, _reconcile_subprocess_callback
app/workflow/preflight_stats.py	One record per preflight-producing call (manual validate, generate-from-prompt, autofix), so we can actually measure — instead of guess — how often workflows come out with basic/repetitive structural errors and which error codes recur most.	ensure_indexes, record_attempt, preflight_stats
app/workflow/prompt_template_store.py	Owner-scoped custom prompt templates plus immutable built-in templates.	template_variables, PromptTemplateRecord, PromptTemplateRecord.public, built_in_templates, ensure_prompt_template_indexes, PromptTemplateStore, PromptTemplateStore.create, PromptTemplateStore.list, PromptTemplateStore.get,

File	Purpose	Major symbols
		PromptTemplateStore.update, PromptTemplateStore.delete
app/workflow/run_chat_store.py	Durable storage for "Ask AI about this run" conversations.	_require_session, ensure_run_chat_indexes, get_run_chat_turns, append_run_chat_turn
app/workflow/run_history.py	Durable workflow-run history.	_is_document_size_error, _externalize_if_large, _inflate_value, _inflate_run_document, _require_session, _node_key, redact_for_history, build_node_input, ensure_indexes, upsert_run, record_node_started, record_node_completed
app/workflow/ subprocess_callback.py	Deliver a finished child run's result to its waiting Subprocess parent.	_select_result, deliver_by_token, deliver_by_child_run_id
app/workflow/ subprocess_launches.py	Subprocess launch correlation — the webhook side of the Subprocess node.	ensure_indexes, find_by_launch_key, reserve_launch, find_by_child_run_id, find_by_token, deliver
deploy/ionos/backup.sh	#!/usr/bin/env bash set -euo pipefail root_dir=/opt/ eurskem release_dir=\${root_dir}/current env_file=\$ {root_dir}/shared/.env.production backup_root=\$ {root_dir}/backups stamp=\$(date -u + %Y%m%d%H%M%S) work_dir=\$(mktemp -d "\$ {backup_root}/.backup-\${stamp}.XXXXXX") archive=\$ {backup_root}/eurskem-\${stamp}.tar.gz stopped=0 cleanup() { if [[\${stopped} -eq 1 && -d \${release_dir}]]; then cd "\${release_dir}" docker compose --env-file "\$ {env_file}" \ -f docker-compose.production.yml up -d fi if [[-d \${work_dir}]]; then rm -rf -- "\${work_dir}" fi } trap cleanup EXIT if [[!	—
deploy/ionos/ cleanup_retired_workflow_artifacts .sh	#!/usr/bin/env bash set -euo pipefail workflows_dir=\${1:? workflow directory is required} if [[!	—
deploy/ionos/deploy_release.sh	#!/usr/bin/env bash set -euo pipefail release_dir=\${1:? release directory is required} release_sha=\${2:?release SHA is required} root_dir=/opt/eurskem shared_dir=\$ {root_dir}/shared env_file=\${shared_dir}/.env.production previous_dir="" if [[!	—
deploy/ionos/init-minio.sh	#!/bin/sh set -eu mc alias set local http://minio:9000 "\$MINIO_ROOT_USER" "\$MINIO_ROOT_PASSWORD" mc mb --ignore-existing "local/\$MINIO_BUCKET" mc admin user add local "\$MINIO_ACCESS_KEY" "\$MINIO_SECRET_KEY" 2>/dev/null true sed "s/ __BUCKET__\$MINIO_BUCKET/g" \ /config/minio-app- policy.json > /tmp	—
deploy/ionos/minio-app-policy.json	{ "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Action": ["s3:GetBucketLocation", "s3:ListBucket"], "Resource": ["arn:aws:s3:::__BUCKET__"] }, { "Effect": "Allow", "Action": ["s3:GetObject", "s3:PutObject",	—

File	Purpose	Major symbols
	"s3:DeleteObject"], "Resource": ["arn:aws:s3:::_BUCKET_/*"]] } }	
deploy/ionos/mongo-init.js	const appDbName = process.env.MONGO_DB "eurskem_ai"; const appDb = db.getSiblingDB(appDbName); if (! appDb.getUser(process.env.MONGO_APP_USERNAME)) { appDb.createUser({ user: process.env.MONGO_APP_USERNAME, pwd: process.env.MONGO_APP_PASSWORD, roles: [{ role: "readWrite", db: appDbName }], }); }	—
deploy/ionos/restore.sh	#!/usr/bin/env bash # Restore an Eurskem AI backup created by backup.sh.	—
deploy/ionos/setup_host.sh	#!/usr/bin/env bash set -euo pipefail if [[\${EUID} -ne 0]]; then echo "Run this once as root on a fresh Ubuntu 24.04 IONOS VPS." >&2 exit 1 fi deploy_user=\$ {DEPLOY_USER:-eurskem-deploy} app_group=eurskem- app app_gid=10001 apt-get update apt-get install -y ca- certificates curl fail2ban rsync ufw install -m 0755 -d /etc/ apt/keyrings curl -fsSL https://download.docker.com/ linux/ubuntu/gpg \ -o /etc/apt/keyrings/docker.asc chmod a+r /etc/apt/keyrings/docker.asc .	—
docker-compose.production.yml	name: eurskem-ai x-logging: &default-logging driver: json- file options: max-size: "10m" max-file: "5" services: caddy: image: caddy:2.10.0-alpine restart: unless- stopped ports: - "80:80" - "443:443" - "443:443/udp" environment: DOMAIN: \${DOMAIN:?Set DOMAIN in .env.production} ACME_EMAIL: \${ACME_EMAIL:?Set ACME_EMAIL in .env.production} MAX_REQUEST_BODY_SIZE: \$ {MAX_REQUEST_BODY_SIZE:-220MB} volumes: - ./ deploy/ionos/Caddyfile:/etc/caddy/Caddyfile:ro - caddy- data:/data - caddy-config:/config depends_on: app: condition: service_healthy frontend: condition: service_healthy networks: - edge logging: *default-logging frontend: build: context: ./ui args: VITE_API_URL: "" restart: unless-stopped networks: - edge healthcheck: test: ["CMD-SHELL", "wget -q --spider http:// 127.0.0.1:8080/healthz"] interval: 20s timeout: 5s retries: 5 start_period: 10s read_only: true tmpfs: - /var/cache/nginx - /var/run - /tmp logging: *default-logging app: build: context: .	—
docker-compose.yml	services: app: build: context: .	—
observability/grafana/dashboards/ platform.json	{ "title": "AWP Platform", "uid": "awp-platform", "schemaVersion": 39, "time": { "from": "now-1h", "to": "now" }, "panels": [{ "title": "Node latency p95 (s) by type", "type": "timeseries", "gridPos": { "h": 8, "w": 12, "x": 0, "y": 0 }, "targets": [{ "expr": "histogram_quantile(0.95, sum(rate(awp	—
observability/grafana/ provisioning/dashboards/awp.yml	apiVersion: 1 providers: - name: awp folder: "" type: file options: path: /etc/grafana/provisioning/dashboards	—
observability/prometheus.yml	global: scrape_interval: 15s scrape_configs: - job_name: "awp-platform" metrics_path: /metrics static_configs: - targets: ["app:8000"]	—
observability/prometheus/ alerts.yml	# Eurskem AI — minimum viable production alert rules.	—

File	Purpose	Major symbols
pyproject.toml	[build-system] requires = ["hatchling"] build-backend = "hatchling.build" [project] name = "agentic-workflow-platform" version = "0.1.0" requires-python = ">=3.11" dependencies = ["fastapi>=0.111.0", "uvicorn[standard]>=0.30.0", "pydantic>=2.7.0", "pydantic-settings>=2.3.0", "structlog>=24.2.0", "python-jose[cryptography]>=3.3.0", "cryptography>=42.0.0", # app/security/passwords.py imports argon2 directly — declared here # explicitly (it previously arrived only transitively via minio).	—
scripts/backfill_knowledge_resources.py	Idempotent legacy -> Knowledge Studio resource backfill.	_load_legacy_collection_configs, _group_manifests_by_owner_and_collection, backfill, main
scripts/build_documentation_pdf.py	Generate the canonical 24-volume codebase documentation portfolio.	Symbol, Route, Node, Model, esc, rel, read, sentence, git_sha, table, markdown, source_files
scripts/build_eu_proposal_copy_bundle.py	Build the copy-ready sourcebook and ZIP for this implementation.	build, main
scripts/build_hitl_editor_bundle.py	Build complete copy/paste files and a path-preserving HITL editor ZIP.	build, main
scripts/build_workflow_file_inputs_bundle.py	Build complete copy/paste files and a path-preserving ZIP.	build, main
scripts/build_workflow_qa_matrix.py	Build a complete, evidence-based QA matrix for every workflow YAML file.	workflow_paths, category, safe_document, input_summary, node_summary, expected_path, load_preflight_reports, build_rows, write_json, write_csv, main
scripts/check_api_routes.py	Fail fast when the Studio frontend and FastAPI route table disagree.	registered_routes, missing_required_routes, main
scripts/debug_compressor.py	Debug the compressor in isolation.	main
scripts/generate_docstrings.py	Generate missing docstrings across the Python codebase.	_words, _phrase, _annotation_text, _default_text, _describe_arg, _summary_for_function, _returns_section, _args_section, _class_summary, _class_doc, _function_doc, _module_doc
scripts/generate_fixtures.py	Generate three synthetic RFP fixtures for the Phase 2 test corpus.	main
scripts/generate_production_env.py	Generate every non-provider secret required by the IONOS Compose stack.	secret, parse_args, main
scripts/generate_reference_workflows.py	Generate 400 hidden, deterministic node-reference workflows.	slug, generate_documents, main
scripts/ingest_samples.py	Repeatable Phase 2 ingestion: seed Weaviate from samples/.	main

File	Purpose	Major symbols
<code>scripts/inspect_weaviate.py</code>	Inspect what's actually indexed in Weaviate.	<code>main</code>
<code>scripts/live_provider_smoke.py</code>	Explicit, low-cost provider smoke test for the protected manual CI job.	<code>smoke_openai</code> , <code>smoke_anthropic</code> , <code>main</code>
<code>scripts/load_test.py</code>	Authenticated 100-concurrent-user release gate with no LLM calls.	<code>main_async</code> , <code>main</code>
<code>scripts/manage_user.py</code>	Create or rotate a Mongo-backed local application user.	<code>parser</code> , <code>run</code> , <code>main</code>
<code>scripts/new_node_type.py</code>	Scaffold one fully standardized workflow node module.	<code>snake_case</code> , <code>main</code>
<code>scripts/openrouter_health_check.py</code>	Live health check for every LLM in OpenRouter's catalog.	<code>HealthReport</code> , <code>CatalogEntry</code> , <code>CatalogEntry.platform_id</code> , <code>CatalogEntry.emits_text</code> , <code>CatalogEntry.is_text_only</code> , <code>CatalogEntry.is_batch_endpoint</code> , <code>CatalogEntry.claims_structured</code> , <code>CatalogEntry.claims_tools</code> , <code>CatalogEntry.claims_reasoning</code> , <code>CatalogEntry.ceiling</code> , <code>CatalogEntry.worst_case_cost_usd</code> , <code>_price</code>
<code>scripts/preflight_workflows.py</code>	Validate workflow YAML files without running nodes or calling an LLM.	<code>_paths</code> , <code>main</code>
<code>scripts/production_preflight.py</code>	Fail-fast validation for the IONOS production environment file.	<code>parse_env</code> , <code>main</code>
<code>scripts/register_biomass.py</code>	Register biomass module.	<code>main</code>
<code>scripts/render_sample_proposal.py</code>	Render the checked-in Horizon proposal sample for visual QA.	<code>main</code>
<code>scripts/render_sample_proposal_docx.py</code>	Render the checked-in Horizon proposal DOCX sample for visual QA.	<code>main</code>
<code>scripts/retag_work_programme.py</code>	Retag work programme module.	—
<code>scripts/seed_collections.py</code>	Seed collections module.	<code>main</code>
<code>scripts/smoke_events.py</code>	Run a workflow with a tracing subscriber.	<code>_MinimalLLM</code> , <code>_MinimalLLM.complete</code> , <code>_MinimalLLM.complete_structured</code> , <code>_MinimalLLM.chat_with_tools</code> , <code>main</code>
<code>scripts/smoke_llm.py</code>	Phase 3 smoke test — LLM gateway end-to-end with fallback routing.	<code>main</code>
<code>scripts/smoke_production.py</code>	Public TLS, readiness, and security-header smoke test.	<code>request</code> , <code>main</code>
<code>scripts/smoke_retrieval.py</code>	Phase 3 smoke test — full retrieval pipeline end-to-end.	<code>main</code>
<code>scripts/smoke_structured.py</code>	Phase 3 smoke test — structured output via the LLM gateway.	<code>Answer</code> , <code>main</code>
<code>scripts/start_snippet_runner_local.sh</code>	<code>#!/usr/bin/env bash set -euo pipefail repo_root=\$(cd "\$(dirname "\${BASH_SOURCE[0]}")/.." && pwd) socket_path=\${SNIPPET_RUNNER_SOCKET_PATH:-/tmp/snippet-runner.sock} pid_file=\${SNIPPET_RUNNER_PID_FILE:-/tmp/eurskem-snippet-runner.pid} log_file=\${SNIPPET_RUNNER_LOG_FILE:-/</code>	—

File	Purpose	Major symbols
	tmp/eurskem-snippet-runner.log} python=\${PYTHON:-"\${repo_root}/.venv/bin/python"} if [[!	
scripts/ verder_knowledge_config.json	{ "owner_scope_id": "ayush", "collection_id": "col_01M0D8GYAD5FBWKTF7GCE1Z16F", "retrieval_profile_id": "retprof_01M0D8GYAQWKK3BSMK0A0384WG" }	—
scripts/verder_process_emails.py	Driver for the Verder Liquids assessment POC.	main
scripts/ verder_reingest_production.py	Re-ingest the 7 Verder pump datasheets into col_01M0D8GYAD5FBWKTF7GCE1Z16F through the REAL production Knowledge Studio pipeline (IngestionJobRunner), so the collection's already-linked Parser/Chunking Profiles (vision_augmented, parent_child) are genuinely exercised and per-file `product` metadata actually lands in Weaviate — none of which happened under the original legacy ingest_file() seeding (scripts/ verder_setup_knowledge.py).	ensure_physical_class_with_pr oduct, main
scripts/verder_setup_knowledge.py	One-time setup for the Verder Liquids assessment POC's product knowledge.	main
scripts/verify_knowledge_studio.py	Live acceptance harness for Knowledge Studio.	Result, Report, Report.record, Report.check, Report.failed, Report.summary, Api, Api.h, Api._send, Api.get, Api.post, Api.jpost
ui/e2e/chat-knowledge- multimodal.spec.ts	React/TypeScript module.	WORKFLOW_NAME, WORKFLOW_PATH, DICTATED_QUESTION, ANSWER, IMAGE_KEY, MODELS, json, nodeRun, installSpeech, Recognition, Utterance, browserWindow
ui/e2e/chat-stress.spec.ts	React/TypeScript module.	ChatRecord, chatYaml, seededRandom, randomChats, random, adjectives, nouns, outputTypes, number, adjective, noun, output
ui/e2e/chat-workspace.spec.ts	React/TypeScript module.	OBJECTIVE, WORKFLOW_ID, SKILL_WORKFLOW_ID, WEB_WORKFLOW_ID, RAG_WORKFLOW_ID, DIAGRAM_WORKFLOW_ID, ANSWER, AMSTERDAM_ANSWER, WEATHER_ANSWER, NO_KNOWLEDGE_ANSWER, UNGROUNDING_ANSWER, EXPERIENCES
ui/e2e/release-gate.spec.ts	React Strict Mode replays the mount effect in development.	SCREENSHOT_ROOT, workflow, id, run, status, created, chatYaml, MockOptions, json, installApi, workflows, runs
ui/src/App.tsx	On a fresh page load the in-memory identity is empty, but the HttpOnly	Mode, StudioRoot, EvalRoot, CostRoot, KnowledgeRoot, ModeLoading, App, openKnowledge

File	Purpose	Major symbols
<code>ui/src/RunCostContext.tsx</code>	React/TypeScript module.	RunCostContext, useSetRunCost
<code>ui/src/api/client.ts</code>	---- auth identity (in-memory) ----	BASE, API, CriterionScore, ExampleResult, RunCandidate, DiscoveredCandidate, ClaimVerificationResult, InternalEvidenceRecord, Scorecard, WorkflowFieldCheck, WorkflowCaseResult, ModelComparisonResult
<code>ui/src/api/knowledge.ts</code>	---- Resource types (mirror app/knowledge/models.py) ----	API, afetch, j, text, payload, detail, jsonHeaders, DocTypeChoice, EmbeddingModelChoice, ResourceStatus, ProfileType, CollectionResource
<code>ui/src/api/types.ts</code>	Mirrors app/runtime/schema.py and app/nodes/registry.py manifest output.	ExecutionKind, NodePreset, NodeAbout, NodeTypeManifest, FieldKind, FieldSpec, DraftInstructionsField, DraftCodeRequest, SchemaPreview, RuleOperator, RuleCondition, RuleConditionGroup
<code>ui/src/components/CopyButton.tsx</code>	React/TypeScript module.	CopyButton, handleCopy
<code>ui/src/components/ModeShell.tsx</code>	React/TypeScript module.	items, ModeShell
<code>ui/src/components/Spinner.tsx</code>	React/TypeScript module.	Spinner
<code>ui/src/components/auth/LoginPage.test.tsx</code>	React/TypeScript module.	user, onLogin
<code>ui/src/components/auth/LoginPage.tsx</code>	Informational banner (e.g.	Props, LoginPage, submittingRef, handleLogin, result
<code>ui/src/components/forms/StartFormRenderer.test.tsx</code>	Mutable holder, written only from onChange (never during render), so	Harness, field, field, field, lineItems, user, user, captured, Tracking, next, productInput, requestKind
<code>ui/src/components/forms/StartFormRenderer.tsx</code>	The one Start-form rendering implementation, used both by the Configure tab's live preview ('interactive={false}') and by RunDialog's real fill-in form ('interactive={true}') — per the form spec's own explicit "reuse the runtime form components for preview, don't build a second rendering implementat	StartFormField, StartFormFileField, StartFormFileValue, labelFor, optionLabel, isRequired, FieldWrapper, isComplex, labelText, TextInput, inputType, EnumOptions
<code>ui/src/components/layout/Sidebar.tsx</code>	React/TypeScript module.	Mode, NAV, SECONDARY, Props, Sidebar, active
<code>ui/src/components/layout/Topbar.tsx</code>	React/TypeScript module.	Mode, Props, TITLES, Topbar
<code>ui/src/components/rag/RagAnswerView.tsx</code>	The RAG process, made understandable without opening internal traces.	RagSourceView, RagRelevantContextItemView, ModelLabelLookup, sourceLink, link, locationLabel,

File	Purpose	Major symbols
		parts, RagAnswerView, link, location
<code>ui/src/components/studio/WorkflowCanvas.tsx</code>	React/TypeScript module.	Node, Edge, getLayoutedElements, g, Props, WorkflowCanvas, rawNodes, rawEdges
<code>ui/src/components/ui/BrandMark.tsx</code>	Shared wordmark lockup used on the splash screen and in the sidebar.	BrandMark, badge
<code>ui/src/components/ui/Icon.tsx</code>	Small, dependency-free icon set — no icon font or external SVG library is	IconName, IconPath, Icon
<code>ui/src/components/workflows/WorkflowProgress.tsx</code>	React/TypeScript module.	WorkflowStatus, WorkflowProgressStep, WorkflowProgress, complete, selected, button
<code>ui/src/components/workflows/WorkflowStepInspector.tsx</code>	React/TypeScript module.	JsonValue, WorkflowStepInspector
<code>ui/src/components/workflows/workflowStatus.test.ts</code>	React/TypeScript module.	—
<code>ui/src/components/workflows/workflowStatus.ts</code>	React/TypeScript module.	WorkflowStatus, WORKFLOW_STATUS_LABEL, normalizeWorkflowStatus, workflowStatusFromNode, workflowStatusFromRun, WORKFLOW_STATUS_GLYPH, WORKFLOW_STATUS_CLASS
<code>ui/src/hooks/useRunEvents.ts</code>	This effect owns one authenticated SSE fetch/reconnect lifecycle.	useRunEvents, controller, run, result, delay
<code>ui/src/hooks/useRunSocket.ts</code>	Reset state because this effect owns a new external WebSocket lifecycle.	useRunSocket, evt
<code>ui/src/index.css</code>	React/TypeScript module.	—
<code>ui/src/main.tsx</code>	Loaded before index.css so Tailwind's utility classes (e.g.	—
<code>ui/src/modes/cost/CostRoot.tsx</code>	React/TypeScript module.	Tab, TABS, usd, Card, SectionTitle, BreakdownList, max, OverviewTab, maxDaily, PricingTab, refresh, timer
<code>ui/src/modes/eval/EvalRoot.tsx</code>	React/TypeScript module.	CRITERIA, NAVY, GOLD, CANDIDATE_MODELS, ScoreBar, pct, color, labelFor, ModelComparisonRow, passPct, passColor, WorkflowModelComparison
<code>ui/src/modes/knowledge/CollectionContext.tsx</code>	Applies a freshly fetched list, keeping the stored choice when it still	CollectionContextValue, CollectionProvider, setCollectionId, apply, keep, next, refresh, value, CollectionBar
<code>ui/src/modes/knowledge/CollectionsPage.tsx</code>	Make the new collection the one every other tab acts on, so the next	CollectionsPage, toggle, precision, create, created, on
<code>ui/src/modes/knowledge/DocumentsIndexesPage.tsx</code>	KnowledgeRoot keys this page by collection, so a switch remounts it with	DocumentsIndexesPage, refresh, activate, activeIndexId

File	Purpose	Major symbols
<code>ui/src/modes/knowledge/IngestionPage.tsx</code>	Job is done — refresh the shared collection so the bar's document and	IngestionPage, timer, choosePreset, next, run, stamp, payloads, parserProfile, chunkingProfile, created, completed, progress
<code>ui/src/modes/knowledge/KnowledgeRoot.test.tsx</code>	First load is empty; after creating, the list contains the new collection.	api, collection, created, list, section, created, cta
<code>ui/src/modes/knowledge/KnowledgeRoot.tsx</code>	Tab content, remounted whenever the selected collection changes so no page can show a previous collection's documents, indexes or results.	Tab, TABS, KnowledgeTabs, showBar, KnowledgeRoot, requested, handler
<code>ui/src/modes/knowledge/PlaygroundPage.tsx</code>	React/TypeScript module.	STRATEGIES, ResultColumn, PlaygroundPage, filters, config, choosePreset, value, next, run, compare, save, StageTimeline
<code>ui/src/modes/knowledge/ProfilesAgentsPage.tsx</code>	The chosen answering model IS the Generation Profile, so pick or mint	ProfilesAgentsPage, refresh, resolveGenerationProfile, existing, created, create, generation, runTest
<code>ui/src/modes/knowledge/TracesPage.tsx</code>	React/TypeScript module.	TracesPage, open
<code>ui/src/modes/knowledge/collectionStore.ts</code>	The one selected Collection, shared by every Knowledge Studio tab.	COLLECTION_STORAGE_KEY, CollectionContextValue, CollectionContext, useCollection, value
<code>ui/src/modes/knowledge/config.ts</code>	React/TypeScript module.	IngestionPresetKey, INGESTION_PRESETS, ingestionProfileRequests, label, comparisonExperiments
<code>ui/src/modes/knowledge/shared.tsx</code>	React/TypeScript module.	ErrorNotice, ResourceId, copy, STATUS_STYLES, Status, style
<code>ui/src/modes/studio/Builder.tsx</code>	Semantic-zoom tiers, with a gap between them so a node's appearance doesn't	Connection, Edge, EdgeChange, Node, NodeChange, ReactFlowInstance, Viewport, ModelRoutingPolicy, NodeExperienceSpec, NoteSpec, WorkflowEdgeData, WorkflowInputSpec
<code>ui/src/modes/studio/BuilderInspector.test.tsx</code>	React/TypeScript module.	user, onTest, onAutofix
<code>ui/src/modes/studio/BuilderInspector.tsx</code>	The node inspector (§18).	BuilderInspectorTab, TABS, NodeActions, BuilderInspector, manifest, previewValues, typeNameOf, upstreamTypes, downstreamTypes, businessLabel, hasChatbotStart, visibleTabs
<code>ui/src/modes/studio/BuilderStart.tsx</code>	React/TypeScript module.	NAME_PATTERN, BuilderStart, trimmed, valid
<code>ui/src/modes/studio/BuilderTestPanel.tsx</code>	React/TypeScript module.	BuilderTestPanel, branches, label
<code>ui/src/modes/studio/Cockpit.tsx</code>		

File	Purpose	Major symbols
	Keep launch-state fields grouped above: they all come from one stable	Edge, Node, ReactFlowInstance, GraphEdge, NodeStatus, nodeTypes, STATUS_BADGE, Cockpit, run, leftPanel, rightPanel, base
<code>ui/src/modes/studio/CockpitNode.tsx</code>	Short duration string (e.g.	CockpitNodeData, STATUS_STYLES, CockpitNodeImpl, s, active, faded, CockpitNode
<code>ui/src/modes/studio/ConfigPanel.tsx</code>	React/TypeScript module.	ConfigPanel, manifest, supportsModelSelection, configuredModel, selectedModel, modelInfo, routingPolicy, updateRoutingPolicy
<code>ui/src/modes/studio/DataMappingPanel.tsx</code>	The Inputs tab — where a step's fields get their values.	PropertySchema, acceptsText, CONFIGURE_TAB_OWNED_FIELDS, humanLabel, leafLabel, leaf, DataMappingPanel, manifest, properties, ownedFields, targetFields, downstream
<code>ui/src/modes/studio/FileInputField.tsx</code>	React/TypeScript module.	FileInputField, extensions, allowed, maximum, addFiles, next, unique, invalid, tooLarge
<code>ui/src/modes/studio/GenerateWorkflowDialog.tsx</code>	A generation that failed its own static/execution check is preflight-	GenerateWorkflowDialog, navigate, generate, res, openInBuilder, workflow, existing, slug
<code>ui/src/modes/studio/GuidedExperiencePanel.tsx</code>	React/TypeScript module.	DEFAULT_STAGE_OPTIONS, PreviewState, cleanExperience, cleaned, GuidedExperiencePanel, experience, fallbackName, stageOptions, custom, update, displayName, purpose
<code>ui/src/modes/studio/HITLPanel.tsx</code>	Guided Run's business-language framing for the same paused node — all	RichEditorValue, DEFAULT_CAPABILITIES, emptyEditorValue, contentAsText, initialReviewContent, first, fallback, formatBytes, errorMessage, HITLPanel, initialContent, jsonAdapter
<code>ui/src/modes/studio/Library.test.tsx</code>	React/TypeScript module.	—
<code>ui/src/modes/studio/Library.tsx</code>	One-time mount fetch, not a sync-to-external-system loop — same	LibraryFilterState, LibrarySortKey, Library, navigate, loadWorkflows, categoryCounts, counts, availableOutputs, outputs, visible, byNav, bySearchAndFilters
	Searchable model picker: the small, curated catalog (llmModels — includes the single "Auto" option) plus a live	ModelSelect, containerRef, timer, handleClick, needle,

File	Purpose	Major symbols
<code>ui/src/modes/studio/ModelSelect.tsx</code>	search over OpenRouter's ~500-model catalog (GET /api/llm/models/openrouter, TTL-cached server-side — see <code>app/llm/openrouter_catalog.py</code>).	<code>filteredCatalog</code> , <code>selectedCatalogModel</code> , <code>selectedLabel</code> , <code>choose</code>
<code>ui/src/modes/studio/NodePalette.test.tsx</code>	React/TypeScript module.	<code>manifest</code> , <code>types</code> , <code>user</code> , <code>user</code> , <code>onAdd</code> , <code>presetTypes</code>
<code>ui/src/modes/studio/NodePalette.tsx</code>	Fixed display order — otherwise groups would shuffle on every reload	<code>CATEGORY_ORDER</code> , <code>HIDDEN_FROM_PALETTE</code> , <code>PALETTE_LABELS</code> , <code>groupByCategory</code> , <code>groups</code> , <code>list</code> , <code>known</code> , <code>unknown</code> , <code>matchesQuery</code> , <code>label</code> , <code>presets</code> , <code>haystack</code>
<code>ui/src/modes/studio/NodeTypeAskAi.tsx</code>	Thin wrapper over the shared <code>AskAiDialog</code> , scoped to one node type — kept as its own component because its two call sites (<code>NodePalette.tsx</code> , <code>cockpit/NodeInspector.tsx</code>) only ever need <code>`typeName`</code> .	<code>starterQuestion</code> , <code>NodeTypeAskAi</code>
<code>ui/src/modes/studio/OutputViewer.tsx</code>	Colour-code each artifact chip by file type so a run with both a PDF and	<code>EXTENSION_STYLE</code> , <code>OutputViewer</code> , <code>navigate</code> , <code>nodeOutputs</code> , <code>workflowInputs</code> , <code>workflowVariables</code> , <code>OutputArtifact</code> , <code>rendererEntries</code> , <code>artifactKeys</code> , <code>artifacts</code> , <code>key</code> , <code>extension</code>
<code>ui/src/modes/studio/PreflightPanel.tsx</code>	React/TypeScript module.	<code>PreflightPanel</code> , <code>errors</code> , <code>warnings</code> , <code>IssueList</code> , <code>clickable</code> , <code>content</code>
<code>ui/src/modes/studio/PromptDraftAssistant.tsx</code>	What this drafting session is for, e.g.	<code>PromptDraftAssistant</code> , <code>bottomRef</code> , <code>ask</code> , <code>trimmed</code> , <code>withQuestion</code> , <code>result</code> , <code>lastAssistantDraft</code>
<code>ui/src/modes/studio/ProposalReview.tsx</code>	Fetching route-owned data is the external synchronization for this effect.	<code>statusClass</code> , <code>ProposalReview</code> , <code>params</code> , <code>concepts</code> , <code>load</code> , <code>addSource</code> , <code>verifyClaims</code> , <code>result</code> , <code>generateConcepts</code> , <code>result</code> , <code>requestApproval</code> , <code>approval</code>
<code>ui/src/modes/studio/RichTextEditor.tsx</code>	Workflow/LLM text is always escaped.	<code>RichEditorValue</code> , <code>EditorCommand</code> , <code>COMMANDS</code> , <code>escapeHtml</code> , <code>plainTextToHtml</code> , <code>RichTextEditor</code> , <code>editorRef</code> , <code>resetTextRef</code> , <code>emitChange</code> , <code>editor</code> , <code>runCommand</code> , <code>insertPlainText</code>
<code>ui/src/modes/studio/RunCandidates.tsx</code>	React/TypeScript module.	<code>DiscoveredCandidate</code> , <code>InternalEvidenceRecord</code> , <code>RunCandidate</code> , <code>foundByLabel</code> , <code>sourceTypeLabel</code> , <code>RunCandidates</code> , <code>navigate</code> , <code>load</code> , <code>res</code> , <code>verifyClaim</code> , <code>res</code> , <code>shown</code>
<code>ui/src/modes/studio/RunDialog.test.tsx</code>	A chatbot-mode Start node declares no <code>`fields:`</code> and no top-level <code>`message`</code>	<code>navigateMock</code> , <code>actual</code> , <code>CHATBOT_WORKFLOW_YAML</code> , <code>renderDialog</code> , <code>field</code>
<code>ui/src/modes/studio/RunDialog.tsx</code>	Fields declared directly on a Start node in <code>`input_form`</code> mode — read from the workflow YAML itself rather than the lossy, collapsed <code>`inputs:`</code> block (every non-file Start field flattens to <code>`type: "json"`</code> there), so the rich widgets	<code>startFormFieldsFrom</code> , <code>parsed</code> , <code>node</code> , <code>config</code> , <code>chatbotStartConfigFrom</code> , <code>parsed</code> , <code>node</code> , <code>config</code> ,

File	Purpose	Major symbols
	(email/currency/date-range/repeating-group/conditional visibility) ar	RunLaunchContext, RunDialog, navigate, startFields
<code>ui/src/modes/studio/RunHistory.tsx</code>	Nothing selected yet — land on the most recent run rather than an	TERMINAL_STATUSES, VALID_TABS, RunHistory, navigate, tabParam, activeTab, selectedNodeId, data, leftPanel, rightPanel, next, selectRun
<code>ui/src/modes/studio/SaveAsDialog.tsx</code>	React/TypeScript module.	SaveAsDialog, onDisplayNameChange, collision, slugValid, canSubmit, submit
<code>ui/src/modes/studio/SchemaForm.tsx</code>	Treat these field names as multiline regardless of schema (the schema	MULTILINE_NAMES, PROMPT_FIELD_NAMES, Schema, SchemaForm, required, next, FieldRenderer, label, askAiDialog, effective, isOptional, resourceKind
<code>ui/src/modes/studio/StudioLayout.tsx</code>	React/TypeScript module.	tabs, StudioLayout, location, chatRoute
<code>ui/src/modes/studio/StudioRoot.test.tsx</code>	React/TypeScript module.	LocationProbe, location, renderRoute, expectRedirect, DeferredScreen
<code>ui/src/modes/studio/StudioRoot.tsx</code>	React/TypeScript module.	Library, Builder, Cockpit, BusinessChat, RunHistory, RunCandidates, ProposalReview, LegacyRunRedirect, StudioRoot
<code>ui/src/modes/studio/VersionHistoryPanel.test.tsx</code>	React/TypeScript module.	YAML, onRestored
<code>ui/src/modes/studio/VersionHistoryPanel.tsx</code>	Reset the preview whenever the selected version changes (including to	formatTimestamp, summarize, parsed, VersionHistoryPanel, currentSummary, previewSummary, selectedVersion, restore, result
<code>ui/src/modes/studio/WorkflowInputsPanel.tsx</code>	React/TypeScript module.	nextInputName, InputNameEditor, valid, duplicate, commit, WorkflowInputsPanel, update, rename, next, remove, next, addInput
<code>ui/src/modes/studio/WorkflowNode.tsx</code>	A step on the canvas.	WorkflowNode, requestedModel, businessLabel, typeLabel, mcpOperation, vertical, border, targetHandle, sourceHandle, deleteButton
<code>ui/src/modes/studio/WorkflowVariablesPanel.tsx</code>	React/TypeScript module.	VariableMap, visibleInputs, valueText, templateReference, VariableValue, rendered, VariableSection, entries, WorkflowVariablesPanel, safeInputs, outputEntries, copy

File	Purpose	Major symbols
<code>ui/src/modes/studio/__tests__/yaml-roundtrip.shipped.test.ts</code>	Vitest provides the CJS <code>__dirname</code> shim even in its jsdom environment	WORKFLOWS_DIR, normalise, roundTrip, original, rebuilt, files, text
<code>ui/src/modes/studio/builder-graph.test.ts</code>	<code>a -> b -> d, a -> c -> d</code>	node, edge, edges, upstream, edges, manifests, nodes, edges, options, sourceClds, nodes, edges
<code>ui/src/modes/studio/builder-graph.ts</code>	Renaming a node must not silently break every other node's template	VariableOption, SchemaShape, schemaFields, properties, renameTokensInText, pattern, renameNodeReferencesInValue, renameNodeReferencesInConfig, EMBEDDED_TOKEN_RE, WHOLE_VALUE_RE, PruneResult, pruneNodeReferencesInValue
<code>ui/src/modes/studio/builder-helpers.ts</code>	Walk a JSON Schema and produce a default value matching it.	generateDefaults, out, d, variants, nonNull, newNodeId, stem, findManifest
<code>ui/src/modes/studio/builder/AITaskConfig.tsx</code>	Configuring the one AI capability.	Config, LANGUAGES, asString, asRecord, AITaskConfig, language, outputFields, task, set, applyPreset, existing
<code>ui/src/modes/studio/builder/AboutPanel.tsx</code>	The About tab (§19).	AboutPanel, about, Entry, Badge, Flag, className
<code>ui/src/modes/studio/builder/AskAiDialog.tsx</code>	The shared Ask AI chat surface (Problem 3) — a small modal thread over POST <code>/api/node-types/ask</code> .	AskAiDialog, askedInitially, bottomRef, ask, trimmed, withQuestion, result
<code>ui/src/modes/studio/builder/ChatPreviewPanel.test.tsx</code>	Technical detail sits behind a collapsed disclosure, not shown by default.	workflow, user, user, user, detail, user
<code>ui/src/modes/studio/builder/ChatPreviewPanel.tsx</code>	Try the workflow as a chatbot, live, from inside the Builder (§1–§46 of the Chat Preview spec).	Turn, ChatOutput, SEND_TIMEOUT_MS, findChatbotStart, node, asChatOutput, humanize, ChatPreviewPanel, startConfig, controllerRef, chatbotName, welcome
<code>ui/src/modes/studio/builder/CloudFileBrowser.test.tsx</code>	Clicking a row in multiple mode does not select immediately.	user, user, onSelect, user, onSelect, fileRow, folderRow, user, onSelect, user, onSelect
<code>ui/src/modes/studio/builder/CloudFileBrowser.tsx</code>	A live folder/search browser for a connected Google Drive or OneDrive account, embedded in the Integration node's configuration panel.	formatBytes, units, formatDate, date, asFileRef, CloudFileBrowser, currentFolderId, loadMore, openFolder, jumpToBreadcrumb, selectable, toggleChecked
<code>ui/src/modes/studio/builder/ConditionGroupEditor.tsx</code>	Nested AND / OR / NOT condition editing.	isGroup, newCondition, newGroup, stripBraces, toReference, coerceValue, trimmed, parsed, valueToText, findContractField, reference, match

File	Purpose	Major symbols
<code>ui/src/modes/studio/builder/ConfigureTab.tsx</code>	A TransformAgent node authored the old way (hand-written <code>prompt_template/ system_prompt/ output_schema</code>) keeps rendering through the generic <code>SchemaForm</code> , unchanged — only a brand-new node gets the Inputs/ Instructions/Outputs editor.	<code>asNonEmptyString</code> , <code>isLegacyTransform</code> , <code>ConfigureTab</code> , <code>config</code> , <code>typeName</code> , <code>transformMode</code> , <code>useNewPromptTemplateEditor</code> , <code>DecisionConfig</code> , <code>rules</code> , <code>defaults</code> , <code>DefaultsEditor</code> , <code>entries</code>
<code>ui/src/modes/studio/builder/DataTransformConfig.tsx</code>	The Transform node's typed operation editor.	<code>Config</code> , <code>OperationKind</code> , <code>Operation</code> , <code>OPERATION_KINDS</code> , <code>OPERATION_LABELS</code> , <code>OPERATION_HINTS</code> , <code>NEEDS_SOURCE</code> , <code>NEEDS_SOURCES</code> , <code>newOperation</code> , <code>operationsOf</code> , <code>PathField</code> , <code>SourcesField</code>
<code>ui/src/modes/studio/builder/EmailConfig.tsx</code>	The Email capability's configuration.	<code>Config</code> , <code>WRITE_OPERATIONS</code> , <code>SENDING_OPERATIONS</code> , <code>asString</code> , <code>Recipient</code> , <code>recipientsOf</code> , <code>useEmailOAuthPopup</code> , <code>onCompleteRef</code> , <code>handleMessage</code> , <code>ConnectMailboxButtons</code> , <code>ConnectionManagement</code> , <code>toggleAllowSend</code>
<code>ui/src/modes/studio/builder/EndAgentConfig.tsx</code>	EndAgent's config editor — what this workflow returns or shows.	<code>Config</code> , <code>EndOutputField</code> , <code>asString</code> , <code>EndAgentConfig</code> , <code>mode</code> , <code>set</code> , <code>MappedValueField</code> , <code>OutputMappingList</code> , <code>outputs</code> , <code>setOutputs</code> , <code>replace</code> , <code>copy</code>
<code>ui/src/modes/studio/builder/ExecutionKindBadge.tsx</code>	Makes the automation boundary visible (§25).	<code>EXECUTION_KINDS</code> , <code>ExecutionKindBadge</code> , <code>meta</code>
<code>ui/src/modes/studio/builder/ExplanationView.tsx</code>	"Why did this happen?" (§24, §47).	<code>ExplanationView</code> , <code>meta</code> , <code>ValueTree</code> , <code>entries</code>
<code>ui/src/modes/studio/builder/ExternalActionConfig.tsx</code>	Configuring a call to a system this deployment has no MCP connection for — a specific URL the author already knows.	<code>Config</code> , <code>METHODS</code> , <code>SAFETY_CLASSES</code> , <code>asString</code> , <code>headerEntries</code> , <code>headers</code> , <code>ExternalActionConfig</code> , <code>actionType</code> , <code>safetyClass</code> , <code>headers</code> , <code>isWrite</code> , <code>setHeaders</code>
<code>ui/src/modes/studio/builder/FieldPicker.tsx</code>	Re-exported for existing call sites (<code>OutputsPanel.tsx</code> , <code>PromptTemplateConfig.tsx</code>)	<code>KIND_LABELS</code> , <code>FIELD_GROUP_COLLAPSE_THRESHOLD</code> , <code>matches</code> , <code>haystack</code> , <code>FieldEntry</code> , <code>depth</code> , <code>leaf</code> , <code>label</code> , <code>ValuePicker</code> , <code>normalised</code> , <code>filterableGroups</code> , <code>groups</code>
<code>ui/src/modes/studio/builder/InfoPopover.tsx</code>	The reusable "❗ Info" trigger for a Builder feature (Problem 1).	<code>InfoPopover</code> , <code>entry</code> , <code>Row</code>
<code>ui/src/modes/studio/builder/IntegrationConfig.test.tsx</code>	React/TypeScript module.	<code>PRESETS</code> , <code>GOOGLE_CONNECTION</code> , <code>user</code> , <code>onChange</code> , <code>user</code> , <code>user</code> , <code>user</code> , <code>onChange</code> , <code>user</code> , <code>onChange</code> , <code>user</code> , <code>onChange</code>

The Integration capability's configuration.

File	Purpose	Major symbols
<code>ui/src/modes/studio/builder/IntegrationConfig.tsx</code>		Config, Provider, Picked, PROVIDER_LABELS, OPERATIONS, asString, isTemplateRef, toldList, useIntegrationOAuthPopup, onCompleteRef, handleMessage, ConnectionManagement
<code>ui/src/modes/studio/builder/JoinConfig.tsx</code>	The Join node's editor (backend type `TextAssemblerAgent` — kept for backward compatibility, see <code>app/nodes/text_assembler.py</code>).	Config, partsOf, SEPARATOR_PRESETS, BranchField, JoinConfig, parts, separator, copy
<code>ui/src/modes/studio/builder/MCPToolConfig.tsx</code>	Re-exported for existing call sites (<code>EmailConfig.tsx</code> , <code>ExternalActionConfig.tsx</code>).	Config, asString, SchemaProperty, propertiesOf, properties, required, primitiveType, declared, MCPToolConfig, serverId, toolName, argumentValues
<code>ui/src/modes/studio/builder/MCPToolPicker.tsx</code>	Add an MCP tool by searching what it does, across every connected system at once — instead of picking a server first and only then discovering what it can do.	Catalog, requiredArgumentNames, required, matchesQuery, haystack, MCPToolPicker, inputRef, perServer, catalog, entries, aRank, bRank
<code>ui/src/modes/studio/builder/NodeSearchPalette.test.tsx</code>	React/TypeScript module.	node, nodes, setup, onSelect, onClose
<code>ui/src/modes/studio/builder/NodeSearchPalette.tsx</code>	Jump straight to a step by name (⌘K / Ctrl-K).	NodeSearchPalette, inputRef, matches, active, next
<code>ui/src/modes/studio/builder/NodeTestPanel.tsx</code>	The Test tab (§21, §22).	exampleValueForType, setDeep, segments, existing, buildSampleInputs, out, buildSampleUpstream, out, nodeSample, NodeTestPanel, supportsModel, runTest
<code>ui/src/modes/studio/builder/NoteNode.tsx</code>	A sticky note on the canvas: a personal annotation the author leaves for themselves, never sent to the backend and never part of the graph (see <code>Builder.tsx</code> 's note handling — it lives in the same react-flow `nodes` array as real steps so drag/select/delete/undo all work for free, but is filtered out everywhere the workflow itself is built or compiled).	NOTE_NODE_TYPE, NOTE_ID_PREFIX, isNoteNodeId, NoteNode
<code>ui/src/modes/studio/builder/OperationBadge.tsx</code>	Shared safety-classification badge — one visual language for every node that touches something outside the platform, not just MCP tools.	OPERATION_STYLES, OPERATION_TITLES, OperationBadge
<code>ui/src/modes/studio/builder/OutputsPanel.tsx</code>	The Outputs tab (§40).	OutputsPanel
<code>ui/src/modes/studio/builder/PromptTemplateConfig.tsx</code>	The Prompt Template editor: Inputs → Instructions → Outputs.	Config, InputRow, INPUT_TYPES, isFileReaderTextField, asString, nextInputName, isValidIdentifier, leafLabel, leaf, destinationKindForType, InputValueEditor, binding
	Write-and-run a Python snippet.	Config, asString, PythonSnippetConfig,

File	Purpose	Major symbols
<code>ui/src/modes/studio/builder/PythonSnippetConfig.tsx</code>		inputFields, outputFields, entries, setInput, next, renameInput, next, value, removeInput
<code>ui/src/modes/studio/builder/RAGAgentConfig.tsx</code>	Configuring a RAG Agent workflow step.	Config, asString, asRecord, RAGAgentConfig, set, ragAgentId, AgentPicker, containerRef, agent, timer, handleClick, choose
<code>ui/src/modes/studio/builder/ResourceSelect.test.tsx</code>	React/TypeScript module.	—
<code>ui/src/modes/studio/builder/ResourceSelect.tsx</code>	Selector for config fields the backend marks with `x-resource` (KnowledgeRetrieval's collection_id / retrieval_profile_id).	Option, ResourceSelect, load, selectClass, missing
<code>ui/src/modes/studio/builder/RouterEditor.test.tsx</code>	React/TypeScript module.	setup, onChange
<code>ui/src/modes/studio/builder/RouterEditor.tsx</code>	The visual router editor.	Config, Branch, asString, branchesOf, raw, toBranchMap, map, RouterEditor, mode, selection, set, ModeCard
<code>ui/src/modes/studio/builder/RuleBuilder.tsx</code>	The no-code rule editor.	newRule, ActionEditor, RuleCard, then, RuleBuilder, copy, RuleAssistant, availableFields, submit, renderRule, lines, verb
<code>ui/src/modes/studio/builder/SQLQueryConfig.tsx</code>	A read-only SQL lookup — the escape hatch for a query the classified business-records tools (Customer Search, Order Search, ...) don't already cover.	Config, asString, SQLQueryConfig, params, entries, setParam, next, renameParam, next, value, removeParam, next
<code>ui/src/modes/studio/builder/SchemaBuilder.tsx</code>	The visual structured-output builder.	KINDS, ITEM_KINDS, newField, nextFieldName, taken, coerceForType, next, FieldRow, hasChildren, showEnum, enumValues, setEnumValues
<code>ui/src/modes/studio/builder/SimulatorPanel.tsx</code>	The workflow simulator (§23, §24, §44).	SimulatorPanel, inputSpecs, stubOutputs, parsed, value, run, StepCard, waiting
<code>ui/src/modes/studio/builder/StageNodes.tsx</code>	The two things the Builder canvas draws that are not workflow steps: the background band marking a stage's column, and the box that stands in for a collapsed stage.	BuilderStageBandNode, BuilderStagePlaceholderNode, preview, remaining
<code>ui/src/modes/studio/builder/StartAgentConfig.tsx</code>	StartAgent's config editor — how this workflow begins.	Config, StartField, CURRENCY_CODES, asString, nextName, taken, nextFileFieldName, currencyPreset, numberUnitPreset, durationPreset, dateRangePreset, addressPreset
<code>ui/src/modes/studio/builder/TemplateTextField.tsx</code>	A free-text field (a prompt, an email body, a document template) that can embed references to earlier steps' data	TOKEN_RE, resolveChipLabel, binding, leaf, inner, buildChip,

File	Purpose	Major symbols
	— shown as chips, not raw <code>`{{node.field}}`</code> syntax, while still serializing to exactly that string underneath.	span, deserialize, fragment, appendText, lines, serialize
<code>ui/src/modes/studio/builder/TestSampleEditor.tsx</code>	Choose-or-write controls for the Test tab's two sample sections — "Sample workflow inputs" and "Sample results from earlier steps" — instead of a blank JSON textarea an author has to hand-build by looking up field names elsewhere in the Builder.	FIELD_GROUP_COLLAPSE_THRESHOLD, getDeep, setDeepImmutable, segments, existing, nested, FieldControl, selected, next, WorkflowInputsEditor, inputs, UpstreamSampleEditor
<code>ui/src/modes/studio/builder/WorkflowInputAgentConfig.tsx</code>	WorkflowInputAgent's config editor.	InputBinding, WorkflowInputAgentConfig, source, fields, sample, binding
<code>ui/src/modes/studio/builder/binding.ts</code>	What a config field's raw stored value actually means to a human.	FieldBinding, REFERENCE_RE, resolveBinding, match, path, resolved, resolveAgainstContract, rest, dot, nodeId, fieldPath, node
<code>ui/src/modes/studio/builder/feature-help.ts</code>	Static, reusable copy for the Builder's "❗ Info" popovers (Problem 1).	FeatureHelpEntry, FEATURE_HELP
<code>ui/src/modes/studio/builder/field-suggest.ts</code>	The field-matching/ranking core behind <code>FieldPicker</code> 's "Recommended" shortlist — split out so a caller that wants a suggestion without opening the picker UI (e.g.	DestinationKind, typeLabel, COMPATIBLE_TYPES, isTypeCompatible, incompatibilityReason, SYNONYMS, tokenize, base, expanded, overlapCount, setB, asContractField
<code>ui/src/modes/studio/builder/formConditions.test.ts</code>	React/TypeScript module.	FormConditionGroup, group, group, group, group, errors, errors, errors, errors, errors
<code>ui/src/modes/studio/builder/formConditions.ts</code>	Client-side twin of <code>app/runtime/rules.py</code> 's <code>evaluate_group</code> , restricted to the operators a Start-form's conditional fields actually need (§16 of the form spec: "Support simple operators...	FormConditionOperator, FormCondition, FormConditionGroup, isGroup, valuesEqual, contains, evaluateCondition, actual, evaluateConditionGroup, results, EMAIL_RE, URL_RE
<code>ui/src/modes/studio/builder/node-search.test.ts</code>	"customer" appears in all three of these; only one starts with it.	node, nodes, many
<code>ui/src/modes/studio/builder/node-search.ts</code>	Matching for the Builder's jump-to-step palette.	NodeSearchMatch, matchNodes, needle, described, businessLabel, typeLabel, ranked, rank, label
<code>ui/src/modes/studio/builder/previewValues.ts</code>	Turns a step's last real test/simulation output into the same dotted-path shape <code>ContractField.path</code> uses, so the picker can show an author what a value actually looked like last time this ran, not just its description.	MAX_PREVIEW_LENGTH, summarize, text, text, flattenOutput, out, walk, buildPreviewValues
<code>ui/src/modes/studio/builder/rule-rendering.test.ts</code>	The rule editor's non-visual logic.	path, rule, rendered, rule, rendered, rendered, rendered, rendered
<code>ui/src/modes/studio/builder/stage-view.test.ts</code>	Wide enough to enclose the stage plus its padding, and tall enough to	node, nodes, edges, stages, bands, bands, onCollapse, result, placeholder, collapsedNodes,

File	Purpose	Major symbols
		collapsedEdges, collapsedStages
<code>ui/src/modes/studio/builder/stage-view.ts</code>	Stage bands and stage collapse for the Builder canvas.	BUILDER_STAGE_BAND_TYPE, , BUILDER_STAGE_PLACEHOLDER_TYPE, SYNTHETIC_PREFIX, BAND_PADDING, BAND_LABEL_HEADROOM, PLACEHOLDER_WIDTH, PLACEHOLDER_HEIGHT, isSyntheticNodeId, bandId, placeholderId, BuilderStageBandData, BuilderStagePlaceholderData
<code>ui/src/modes/studio/builder/test-fixtures.ts</code>	Node-type manifest fixtures for tests.	manifestFixture
<code>ui/src/modes/studio/business-chat/AgentActivityCard.tsx</code>	React/TypeScript module.	statusGlyph, initials, AgentActivityCard, running
<code>ui/src/modes/studio/business-chat/AgentActivityGroup.test.tsx</code>	React/TypeScript module.	activities, onSelectNode
<code>ui/src/modes/studio/business-chat/AgentActivityGroup.tsx</code>	React/TypeScript module.	statusGlyph, activityHeadline, reused, AgentActivityGroup, visible, toolCount, duration, current
<code>ui/src/modes/studio/business-chat/BusinessChat.tsx</code>	Business Chat: a published workflow experienced as a conversation.	LocalChatRecord, CitationTarget, CreateArtifactKind, WorkspaceNote, WorkspaceSource, SlashCommand, BrowserSpeechRecognition, AssistantSegment, AgentActivity, WorkflowChatMeta, DurableChatMessage, ChatMessage
<code>ui/src/modes/studio/business-chat/ChatInterventionCard.test.tsx</code>	React/TypeScript module.	message, user, user, editor, user, ref, user
<code>ui/src/modes/studio/business-chat/ChatInterventionCard.tsx</code>	React/TypeScript module.	InterventionMessage, DEFAULT_CAPABILITIES, reviewValue, ReviewFact, ChatInterventionCard, adapter, initialText, canApprove, canEdit, canReject, dirty, overLimit
<code>ui/src/modes/studio/business-chat/ChatNodeInspector.tsx</code>	React/TypeScript module.	ChatNodeInspector
<code>ui/src/modes/studio/business-chat/ChatSessionsPanel.test.tsx</code>	React/TypeScript module.	chat, onOpen, onNew
<code>ui/src/modes/studio/business-chat/ChatSessionsPanel.tsx</code>	React/TypeScript module.	ChatSessionsPanel
<code>ui/src/modes/studio/business-chat/ChatWorkspaceOverlays.test.tsx</code>	React/TypeScript module.	onAddUrls, connection, onImportDrive, onClose, trigger,

File	Purpose	Major symbols
		dialog, close, technical, chat, onOpen, onOpen, onOpen
ui/src/modes/studio/business-chat/ChatWorkspaceOverlays.tsx	React/TypeScript module.	ExistingChatWorkflow, Drawer, dialogRef, restoreFocusRef, dialog, first, focusable, first, ChatHistoryDrawer, ExistingWorkflowDrawer, normalized, visible
ui/src/modes/studio/business-chat/ChatWorkspaceShell.tsx	React/TypeScript module.	ChatWorkspacePanel, clamp, ChatWorkspaceShell, dragRef, leftWidth, rightWidth, style, beginResize, startWidth, resize, drag, delta
ui/src/modes/studio/business-chat/ComposerMenu.test.tsx	React/TypeScript module.	items, onClose, analyze, research, onClose, skill, createItem
ui/src/modes/studio/business-chat/ComposerMenu.tsx	React/TypeScript module.	ComposerMenuItem, ComposerMenu, rootRef, restoreFocusRef, closeOutside, moveFocus, buttons, current
ui/src/modes/studio/business-chat/NotebookSourcesPanel.test.tsx	React/TypeScript module.	sources, onToggle, onShowUsage, onToggle, onRemoveSource
ui/src/modes/studio/business-chat/NotebookSourcesPanel.tsx	React/TypeScript module.	SourceFilter, sourceIcon, NotebookSourcesPanel, visible, matchesQuery, matchesFilter, selected
ui/src/modes/studio/business-chat/PromptTemplateLibrary.test.tsx	React/TypeScript module.	template, user, onInsert, custom, user, copy, user
ui/src/modes/studio/business-chat/PromptTemplateLibrary.tsx	React/TypeScript module.	CATEGORIES, PromptTemplateLibrary, visible, duplicate, favorite, remove, save, result
ui/src/modes/studio/business-chat/RunControlBar.tsx	React/TypeScript module.	RunControlBar, state
ui/src/modes/studio/business-chat/SessionAuditPanel.test.tsx	React/TypeScript module.	run, onTabChange, events, onSelectNode, source, activity
ui/src/modes/studio/business-chat/SessionAuditPanel.tsx	React/TypeScript module.	SessionTab, statusGlyph, formatDuration, SessionAuditPanel, tab, selectedSources, accessedSources, models, filteredAudit, trace, traceText, downloadTrace
ui/src/modes/studio/business-chat/SourceFirstChatHome.tsx	React/TypeScript module.	CreateArtifactKind, WorkspaceNote, WorkspaceSource, LocalChatRecord, SUGGESTIONS, SourceFirstChatHome, navigate, workspaceId, openChat, newChat, chat, openWorkflowPicker

File	Purpose	Major symbols
<code>ui/src/modes/studio/business-chat/WorkflowContextPanel.tsx</code>	React/TypeScript module.	stageStatus, WorkflowContextPanel, businessStages, label, status
<code>ui/src/modes/studio/business-chat/businessChatModel.test.ts</code>	Node-side test reading the repo's reference corpus (same convention as	WorkflowChatMeta, REFERENCE_DIR, fileRef, metaDefaults, yamlText, meta, yamlText, meta, meta, chatbotMeta, meta, result
<code>ui/src/modes/studio/business-chat/businessChatModel.ts</code>	Pure model layer for the Business Chat experience.	ChatOutput, GenericChatExperienceId, GENERIC_CHAT_EXPERIENCE_S, ChatFormField, WorkflowChatMeta, WorkflowChatNode, businessActivityLabel, AgentActivity, chatMetaFromYaml, meta, nodes, upstream
<code>ui/src/modes/studio/business-chat/chatEnhancements.test.ts</code>	React/TypeScript module.	table, research
<code>ui/src/modes/studio/business-chat/chatEnhancements.ts</code>	React/TypeScript module.	ResponseFormat, WritingStyle, SlashCommand, SLASH_COMMANDS, matchingSlashCommands, extractTemplateVariables, found, renderPromptTemplate, classifyResponseFormat, text, numericSignals, formatHint
<code>ui/src/modes/studio/business-chat/chatOutputs.test.ts</code>	React/TypeScript module.	fileRef, answer, result, result, renderer, reference, structured, result, result
<code>ui/src/modes/studio/business-chat/chatOutputs.ts</code>	Artifacts emitted by the same output object are siblings.	EMPTY_CHAT_RESULT, ChatArtifactKind, ChatArtifact, ChatOutput, ChatOutputRun, IMAGE_EXTENSIONS, ARTIFACT_EXTENSIONS, ARTIFACT_FIELDS, NON_VISIBLE_KEYS, isWorkflowFileReference, record, extensionOf
<code>ui/src/modes/studio/business-chat/chatTranscript.test.ts</code>	React/TypeScript module.	summary, local, serialized, restored, local, serialized, malformed, restored
<code>ui/src/modes/studio/business-chat/chatTranscript.ts</code>	React/TypeScript module.	DurableChatMessage, SUMMARY_MAX_MESSAGES, SUMMARY_MAX_MESSAGE_CHARS, SUMMARY_MAX_CHARS, boundedConversationSummary, lines, answer, record, answerFromLegacyTechnicalText, trimmed, match, raw
<code>ui/src/modes/studio/business-chat/chatWorkspace.css</code>	React/TypeScript module.	—
	React/TypeScript module.	—

File	Purpose	Major symbols
<code>ui/src/modes/studio/business-chat/chatWorkspaceModel.test.ts</code>		
<code>ui/src/modes/studio/business-chat/chatWorkspaceModel.ts</code>	React/TypeScript module.	WorkspaceSource, collectionAsSource, WorkspaceNote, CitationTarget, CreateArtifactKind, CREATE_OPTIONS, documentsAsSources, uploadsAsSources, driveFilesAsSources, cloudFile, httpUrlsInText, matches
<code>ui/src/modes/studio/business-chat/chatWorkspaceStorage.test.ts</code>	React/TypeScript module.	stubStorage, values, note, sources, sources, first, second, first, second
<code>ui/src/modes/studio/business-chat/chatWorkspaceStorage.ts</code>	Browser-local notes are optional; storage failures must not block Chat.	PREFIX, PREFERENCES_KEY, SOURCE_HANDOFF_PREFIX, SOURCES_PREFIX, CHAT_HISTORY_KEY, LocalChatRecord, LocalChatHistory, emptyHistory, loadLocalChatHistory, value, parsed, saveLocalChatHistory
<code>ui/src/modes/studio/business-chat/observeChatRun.test.ts</code>	React/TypeScript module.	stream, events
<code>ui/src/modes/studio/business-chat/observeChatRun.ts</code>	Observe one attempt until a terminal run event.	ChatRunObserverOptions, observeChatRun, result, error, delay
<code>ui/src/modes/studio/business-chat/runControls.test.ts</code>	React/TypeScript module.	—
<code>ui/src/modes/studio/business-chat/runControls.ts</code>	React/TypeScript module.	RunControlAction, RunControlInput, RunControlState, runControlState, busy, needsReview, attemptLabel, label
<code>ui/src/modes/studio/business-chat/sessionProjection.test.ts</code>	React/TypeScript module.	audit, source, activities, trace
<code>ui/src/modes/studio/business-chat/sessionProjection.ts</code>	React/TypeScript module.	AuditFilter, APPROVAL_EVENTS, auditEventLabel, labels, filterAuditEvents, auditPayloadSummary, payload, baseActivitySourceTitle, sourceActivityNodeIds, sourceTitle, matches, buildSessionTrace
<code>ui/src/modes/studio/business-chat/speech.test.ts</code>	React/TypeScript module.	Recognition, standard, prefixed, speak, cancel, Utterance, scope
<code>ui/src/modes/studio/business-chat/speech.ts</code>	React/TypeScript module.	SpeechRecognitionResult, RecognitionAlternative, RecognitionResultLike, RecognitionEventLike,

File	Purpose	Major symbols
		BrowserSpeechRecognition, RecognitionConstructor, recognitionConstructor, candidate, speechRecognitionSupported, speechSynthesisSupported, collectRecognitionText, result
ui/src/modes/studio/cockpit-state.test.ts	join is still reachable via `left`, so it must NOT be skipped even	NodeStatus, state, state, nodes, edges, states, result, edges, states, outputs, result, edges
ui/src/modes/studio/cockpit-state.ts	'pending' is the internal value for the user-facing "Waiting" status —	NodeStatus, RunStatus, CockpitState, initialCockpitState, nodeStates, deriveCockpitState, s, GraphNode, GraphEdge, TERMINAL_DEAD_END, DECIDED, chosenRoute
ui/src/modes/studio/cockpit/JsonTree.tsx	A small recursive, collapsible, searchable JSON tree — no external	JsonValue, valueMatches, lower, Leaf, JsonNode, isCollapsible, forcedOpen, expanded, entries, visibleEntries, isArray, summary
ui/src/modes/studio/cockpit/NodeInspector.tsx	Only meaningful once the run has stopped changing under it — while a	SelectedNodeInfo, TabKey, TABS, durationLabel, OverviewTabContent, ts, askAiAvailable, InputTabContent, MetadataTabContent, selections, costLabel, AICostCallCard
ui/src/modes/studio/cockpit/OverviewPanel.tsx	§31/§33: workflow-level AI cost, broken down by node and (when present) by	OverviewNode, FILTER_OPTIONS, STATUS_TO_FILTER, costLabel, AICostSummary, nodeRows, stageRows, elapsedLabel, end, seconds, minutes, remSeconds
ui/src/modes/studio/cockpit/ResizeHandle.tsx	React/TypeScript module.	ResizeHandle
ui/src/modes/studio/cockpit/StageBandNode.tsx	A background rectangle + label marking one stage's column, rendered behind the real nodes (given a lower zIndex on the node itself — see where these are constructed) so parallel branches read as visually grouped without needing a real ReactFlow parent/child node hierarchy.	STAGE_BAND_TYPE, StageBandData, StageBandNode
ui/src/modes/studio/cockpit/StagePlaceholderNode.tsx	Stands in for a collapsed stage's member nodes — click to expand again.	StagePlaceholderNode
ui/src/modes/studio/cockpit/VirtualList.tsx	A small fixed-row-height windowed list.	OVERSCAN, VirtualList, containerRef, el, observer, entry, effectiveHeight, visibleCount, startIndex, endIndex, totalHeight, offsetY
ui/src/modes/studio/cockpit/graph-collapse.test.ts	a -> placeholder (deduped from two edges), placeholder -> d (deduped)	

File	Purpose	Major symbols
		node, nodes, edges, stages, result, result, ids, placeholder, sources
<code>ui/src/modes/studio/cockpit/graph-collapse.ts</code>	Background band nodes for every EXPANDED stage (a collapsed stage is already fully represented by its one placeholder node, so it gets no band).	STAGE_PLACEHOLDER_TYPE, BAND_PADDING, BAND_LABEL_HEADROOM, buildStageBandNodes, width, height, StagePlaceholderData, placeholderId, applyStageCollapse, nodeById, stageForNode, collapsedStages
<code>ui/src/modes/studio/cockpit/node-render.test.ts</code>	React/TypeScript module.	makeNodeRun, result, result
<code>ui/src/modes/studio/cockpit/node-render.tsx</code>	Shared per-node rendering helpers, originally local to RunHistory.tsx and	STATUS_PILL, StatusPill, TYPE_STYLE, TYPE_FALLBACK, typeStyle, NODE_RUN_STATUS_MAP, historicalNodeStatus, run, shortDuration, clock, d, renderValue
<code>ui/src/modes/studio/cockpit/tabs/ErrorsTab.tsx</code>	React/TypeScript module.	ErrorsTab, suggestion, canRetry
<code>ui/src/modes/studio/cockpit/tabs/LogsTab.tsx</code>	React/TypeScript module.	LogLine, buildLogLines, lines, LogsTab, lines
<code>ui/src/modes/studio/cockpit/tabs/OutputTab.tsx</code>	React/TypeScript module.	RagAgentOutput, download, blob, url, a, OutputTab, artifact, key, isImage, isFile, text, isStructured
<code>ui/src/modes/studio/cockpit/useCockpitRun.ts</code>	Extracted, behavior-preserving, from the original Cockpit.tsx: owns the	Gate, Finished, CockpitNavState, liveRunNodeStatus, nodeRunStatus, useCockpitRun, location, navigate, setRunCost, applyResumeResult, interrupt, v
<code>ui/src/modes/studio/cockpit/useResizablePanel.ts</code>	Drag-to-resize a panel's width, persisted per storageKey so it survives reloads/navigation.	useResizablePanel, stored, dragStart, onPointerDown, onPointerMove, delta, signedDelta, next, onPointerUp
<code>ui/src/modes/studio/file-artifact.ts</code>	Detect a downloadable object-store key inside a node's output.	fileKey, obj, v, artifactLabel, ext, obj, parts, size, pages, estimated
<code>ui/src/modes/studio/fileInputUtils.ts</code>	React/TypeScript module.	isFileReferenceLike, fileReferencesFrom, FALLBACK_CATEGORIES, FALLBACK_FILE_CAPABILITIES, extensionOf, dot, formatBytes, acceptedExtensions, accepts
<code>ui/src/modes/studio/flow-layout.ts</code>	How close two nodes' x-positions need to be (in LR layout) to count as	NODE_WIDTH, NODE_HEIGHT, STAGE_TOLERANCE, Stage, LayoutResult, layoutFlow,

File	Purpose	Major symbols
		graph, laidOut, point, width, height, stageLabel
<code>ui/src/modes/studio/graph-export.test.ts</code>	Rightmost node ends at 880; plus padding either side.	node, nodes, edges, image, withExperience, banded, hostile, box, path, across, stacked, wide
<code>ui/src/modes/studio/graph-export.ts</code>	Renders a workflow graph to a standalone SVG document, and rasterises that to PNG.	DEFAULT_NODE_WIDTH, DEFAULT_NODE_HEIGHT, MIN_CARD_HEIGHT, PADDING, HEADER_HEIGHT, BAND_LABEL_HEADROOM, COLORS, BadgeStyle, KIND_BADGES, NEUTRAL_BADGE, READ_BADGE, WRITE_BADGE
<code>ui/src/modes/studio/graph-navigation.test.ts</code>	Both branches are one edge away; 'approve' is nearer classify's own row.	node, edge, nodes, edges, withNearbyOtherColumn, orphaned
<code>ui/src/modes/studio/graph-navigation.ts</code>	Keyboard travel across a workflow graph.	NavDirection, ARROW_DIRECTIONS, DEFAULT_NODE_WIDTH, DEFAULT_NODE_HEIGHT, COLUMN_TOLERANCE, OFF_AXIS_PENALTY, NodeCentre, nodeCentre, firstNodeId, centres, connected, ids
<code>ui/src/modes/studio/guided/runtime-model.test.ts</code>	React/TypeScript module.	workflow, model, byId, model, model, createStage, model, createStage, model, model, withGoal, withoutGoal
<code>ui/src/modes/studio/guided/runtime-model.ts</code>	Final packaging and checking terms are more specific than generic words	GuidedStageState, GuidedStep, GuidedStage, GuidedRuntimeModel, GuidedArtifact, DEFAULT_STAGES, TERMINAL_NODE_STATES, GUIDED_STATUS_LABEL, humanizeIdentifier, spaced, compactText, text
<code>ui/src/modes/studio/hitl-plain-text.test.ts</code>	React/TypeScript module.	adapter, adapter, adapter, adapter, adapter
<code>ui/src/modes/studio/hitl-plain-text.ts</code>	Adapts a JSON review document to a plain-text Business Chat editor.	JsonPathPart, PlainTextJsonAdapter, PREFERRED_TEXT_KEYS, cloneJson, valueAtPath, replaceAtPath, copy, stringPaths, preferredStringPath, paths, match, readableText
<code>ui/src/modes/studio/library/CategoryNav.tsx</code>	Only meaningful below the 900px breakpoint, where this nav becomes an	LibraryNavSelection, CategoryNav, count
<code>ui/src/modes/studio/library/ConfirmDeleteDialog.tsx</code>	React/TypeScript module.	ConfirmDeleteDialog, title, confirm
	React/TypeScript module.	

File	Purpose	Major symbols
<code>ui/src/modes/studio/library/ImportWorkflowDialog.tsx</code>		NAME_PATTERN, ImportWorkflowDialog, nameValid, submit
<code>ui/src/modes/studio/library/LibraryToolbar.tsx</code>	React/TypeScript module.	LibraryDurationBucket, LibraryFilterState, LibrarySortKey, LibraryViewModel, STATUS_OPTIONS, STATUS_LABEL, DURATION_OPTIONS, toggleInSet, next, LibraryToolbar, active
<code>ui/src/modes/studio/library/PrepareAndRunPanel.tsx</code>	The full service-probing check (check_services=true) — same gate	Step, PrepareAndRunPanel, navigate, checkReadinessNow, report, startNow, runId, title, inputCount, needsInputsStep, reviewCount, blocked
<code>ui/src/modes/studio/library/WorkflowCard.tsx</code>	React/TypeScript module.	STATUS_LABEL, durationText, duration, WorkflowCard, title, summary, status, reviewCount, outputs, menuRef, onPointerDown
<code>ui/src/modes/studio/library/WorkflowDetailsPanel.tsx</code>	Structural data (YAML + fresh library/readiness) loads eagerly — it's	DetailsTab, TABS, WorkflowDetailsPanel, title, library, readiness, activeTabContent
<code>ui/src/modes/studio/library/categories.test.ts</code>	Falls through to keyword inference since no declared tag matched.	metadata, categories, categories, categories, categories
<code>ui/src/modes/studio/library/categories.ts</code>	Fixed purpose taxonomy for Library discovery.	LibraryCategoryId, LibraryCategory, LIBRARY_CATEGORIES, CATEGORY_LABEL, slugTold, categoriesForWorkflow, declaredTags, bySlug, matched, searchable, inferred
<code>ui/src/modes/studio/library/filters.test.ts</code>	React/TypeScript module.	readiness, library, workflow, wf, wf, wf, wf, filters, wf, filters, wf, filters
<code>ui/src/modes/studio/library/filters.ts</code>	React/TypeScript module.	LibraryDurationBucket, DURATION_BUCKET_LABEL, LibraryFilterState, emptyFilterState, hasActiveFilters, matchesSearch, trimmed, haystack, durationBucket, duration, minutes, matchesFilters
<code>ui/src/modes/studio/library/localState.test.ts</code>	This project's vitest runs in the default 'node' test environment (no	stubLocalStorage, store
<code>ui/src/modes/studio/library/localState.ts</code>	Favorites and "recently used"/"continue where you left off" are stored in	FAVORITES_KEY, RECENT_KEY, MAX_RECENT, readJsonArray, raw, parsed, writeJsonArray, getFavorites,

File	Purpose	Major symbols
		isFavorite, toggleFavorite, current, getRecentlyOpened
ui/src/modes/studio/library/readiness.test.ts	React/TypeScript module.	report, readiness, readiness, readiness, readiness
ui/src/modes/studio/library/readiness.ts	Mirrors app/workflow/library_metadata.py's readiness_summary exactly —	readinessFromPreflight, errors, warnings, level, items, READINESS_LABEL
ui/src/modes/studio/library/tabs/EvidenceTab.tsx	Deterministic node-type → evidence-source mapping.	EVIDENCE_SOURCE_BY_NODE_TYPE, ALL_SOURCE_LABELS, EvidenceTab, presentTypes, supported
ui/src/modes/studio/library/tabs/OverviewTab.tsx	React/TypeScript module.	STATUS_LABEL, durationText, duration, OverviewTab, model
ui/src/modes/studio/library/tabs/RunsTab.tsx	React/TypeScript module.	formatDuration, minutes, remaining, relativeDate, RunsTab
ui/src/modes/studio/library/tabs/StagesTab.tsx	No live run — this is a structural preview of the same stage/step model	StagesTab, model, toggle, next, steps, reviewSteps, isOpen
ui/src/modes/studio/library/tabs/TechnicalTab.tsx	React/TypeScript module.	TechnicalTab
ui/src/modes/studio/library/tabs/VersionsTab.tsx	React/TypeScript module.	VersionsTab
ui/src/modes/studio/library/tabs/WhatItProducesTab.tsx	React/TypeScript module.	OutputNodeRef, outputNodeIds, output, WhatItProducesTab, nodesById, finalOutputs, node, experience
ui/src/modes/studio/library/tabs/WhatYouNeedTab.tsx	React/TypeScript module.	TYPE_LABEL, WhatYouNeedTab, entries, required, optional
ui/src/modes/studio/node-presentation.ts	How a step is named on the canvas.	TYPE_LABELS, INTEGRATION_PROVIDER_LABELS, humanizeToolName, nodeTypeLabel, base, serverId, tool, provider, count, fileCount, total, count
ui/src/modes/studio/run-history/AskAiPanel.test.tsx	jsdom doesn't implement scrollIntoView; AskAiPanel calls it to keep the	user
ui/src/modes/studio/run-history/AskAiPanel.tsx	Lets a caller open the panel already holding the question the person	formatTime, Avatar, AskAiPanel, bottomRef, inputRef, ask, trimmed, result
ui/src/modes/studio/run-history/RunListPanel.tsx	React/TypeScript module.	GroupBy, ROW_HEIGHT, MAX_GROUP_LIST_HEIGHT, STATUS_OPTIONS, SORT_OPTIONS, GROUP_OPTIONS, shortRunId, progressFraction, RunRow, RunListPanel, filters, activeFilterCount
	React/TypeScript module.	WorkspaceTab, TABS, durationLabel,

File	Purpose	Major symbols
<code>ui/src/modes/studio/run-history/RunWorkspace.tsx</code>		<code>buildReusableInputsJson</code> , <code>nodeRunById</code> , <code>nodeIds</code> , <code>nodeOutputs</code> , <code>downloadJson</code> , <code>blob</code> , <code>url</code> , <code>a</code> , <code>RunWorkspace</code>
<code>ui/src/modes/studio/run-history/run-list-utils.test.ts</code>	React/TypeScript module.	<code>makeRun</code> , <code>run</code> , <code>runs</code> , <code>result</code> , <code>result</code> , <code>result</code> , <code>runs</code> , <code>copy</code> , <code>now</code> , <code>runs</code> , <code>groups</code>
<code>ui/src/modes/studio/run-history/run-list-utils.ts</code>	Display label for a <code>RunStatus</code> — distinct from cockpit-state's <code>NodeStatus</code> labels.	<code>RUN_STATUS_LABEL</code> , <code>SortKey</code> , <code>GroupBy</code> , <code>RunFilters</code> , <code>runTimestamp</code> , <code>matchesSearch</code> , <code>needle</code> , <code>haystack</code> , <code>runHasErrors</code> , <code>filterRuns</code> , <code>ts</code> , <code>sortRuns</code>
<code>ui/src/modes/studio/run-history/tabs/ErrorsTab.tsx</code>	React/TypeScript module.	<code>ErrorCard</code> , <code>suggestion</code> , <code>ErrorsTab</code> , <code>nodeRuns</code> , <code>failedNodes</code> , <code>hasRunLevelError</code>
<code>ui/src/modes/studio/run-history/tabs/InputsTab.tsx</code>	React/TypeScript module.	<code>isEmptyValue</code> , <code>downloadJson</code> , <code>blob</code> , <code>url</code> , <code>a</code> , <code>InputsTab</code> , <code>entries</code> , <code>toggleField</code> , <code>next</code> , <code>expandAll</code> , <code>collapseAll</code> , <code>empty</code>
<code>ui/src/modes/studio/run-history/tabs/NodesTab.tsx</code>	Nodes whose recorded start times are within this many seconds of each other are considered to have run concurrently, not sequentially.	<code>Row</code> , <code>TERMINAL_STATUSES</code> , <code>buildRows</code> , <code>nodeRunById</code> , <code>isTerminal</code> , <code>extraIds</code> , <code>allIds</code> , <code>nodeRun</code> , <code>CONCURRENCY_WINDOW_S</code> , <code>concurrentGroupKey</code> , <code>started</code> , <code>other</code>
<code>ui/src/modes/studio/run-history/tabs/OutputsTab.tsx</code>	React/TypeScript module.	<code>OutputCard</code> , <code>buildCards</code> , <code>nodeRunById</code> , <code>ids</code> , <code>OutputCardView</code> , <code>ts</code> , <code>artifact</code> , <code>preview</code> , <code>OutputsTab</code> , <code>cards</code> , <code>filtered</code> , <code>key</code>
<code>ui/src/modes/studio/run-history/tabs/OverviewTab.tsx</code>	React/TypeScript module.	<code>TERMINAL_STATUSES</code> , <code>allNodeIds</code> , <code>FailureCard</code> , <code>failedNode</code> , <code>totalCompleted</code> , <code>total</code> , <code>reused</code> , <code>rawError</code> , <code>suggestion</code> , <code>OverviewTab</code> , <code>isTerminal</code> , <code>nodeIds</code>
<code>ui/src/modes/studio/run-history/tabs/TimelineTab.tsx</code>	React/TypeScript module.	<code>EVENT_TYPE_LABEL</code> , <code>rowNodeid</code> , <code>rowTs</code> , <code>rowMatchesQuery</code> , <code>needle</code> , <code>haystack</code> , <code>TimelineTab</code> , <code>merged</code> , <code>filteredMerged</code> , <code>filteredRaw</code> , <code>filtered</code> , <code>isFailure</code>
<code>ui/src/modes/studio/run-history/timeline-utils.test.ts</code>	React/TypeScript module.	<code>ev</code> , <code>rows</code> , <code>rows</code> , <code>rows</code> , <code>rows</code> , <code>rows</code> , <code>rows</code>
<code>ui/src/modes/studio/run-history/timeline-utils.ts</code>	One coherent row per node execution (start+end merged) or human decision, instead of the raw audit feed's separate "Node started"/"Node completed" rows.	<code>TimelineRow</code> , <code>END_EVENT_TYPES</code> , <code>nodeStatusForEventType</code> , <code>mergeNodeEvents</code> , <code>ordered</code> , <code>pendingStarts</code> , <code>rows</code> , <code>list</code> , <code>list</code> , <code>start</code> , <code>reason</code> , <code>reason</code>
<code>ui/src/modes/studio/run-history/useRunHistoryData.test.tsx</code>	React/TypeScript module.	<code>wrapper</code> , <code>completedRun</code>

File	Purpose	Major symbols
<code>ui/src/modes/studio/run-history/useRunHistoryData.ts</code>	Extracted, behavior-preserving, from the original <code>RunHistory.tsx</code> : owns	<code>DELETE_MIN_RUNNING_AGE_SECONDS</code> , <code>deleteBlockedReason</code> , <code>ageSeconds</code> , <code>remaining</code> , <code>hoursLeft</code> , <code>TERMINAL_STATUSES</code> , <code>RUN_LIST_LIMIT</code> , <code>useRunHistoryData</code> , <code>navigate</code> , <code>schedule</code> , <code>load</code> , <code>data</code>
<code>ui/src/modes/studio/workflow-naming.ts</code>	Mirrors the backend's defensive check in <code>app/api/workflows.py:save_workflow</code> —	<code>slugify</code> , <code>isValidSlug</code> , <code>uniqueSlug</code> , <code>root</code>
<code>ui/src/modes/studio/yaml-bridge.test.ts</code>	Deriving runnable inputs from the Start node is a read-time concern	<code>YamlWorkflow</code> , <code>roundTrip</code> , <code>workflow</code> , <code>result</code> , <code>workflow</code> , <code>result</code> , <code>workflow</code> , <code>result</code> , <code>node</code> , <code>workflow</code> , <code>result</code> , <code>workflow</code>
<code>ui/src/modes/studio/yaml-bridge.ts</code>	What's in a YAML workflow, parsed shape	<code>WorkflowInputSpec</code> , <code>FILE_CATEGORIES</code> , <code>valueForJsonInput</code> , <code>ModelRoutingPolicy</code> , <code>GuidedVisibility</code> , <code>GuidedStageSpec</code> , <code>WorkflowExperienceSpec</code> , <code>NodeExperienceSpec</code> , <code>YamlWorkflowNode</code> , <code>NoteSpec</code> , <code>YamlWorkflow</code> , <code>WorkflowNodeData</code>
<code>ui/src/styles/globals.css</code>	React/TypeScript module.	—
<code>ui/src/test/setup.ts</code>	<code>vite.config.ts</code> runs <code>vitest</code> without <code>`globals: true`</code> , so React Testing	—